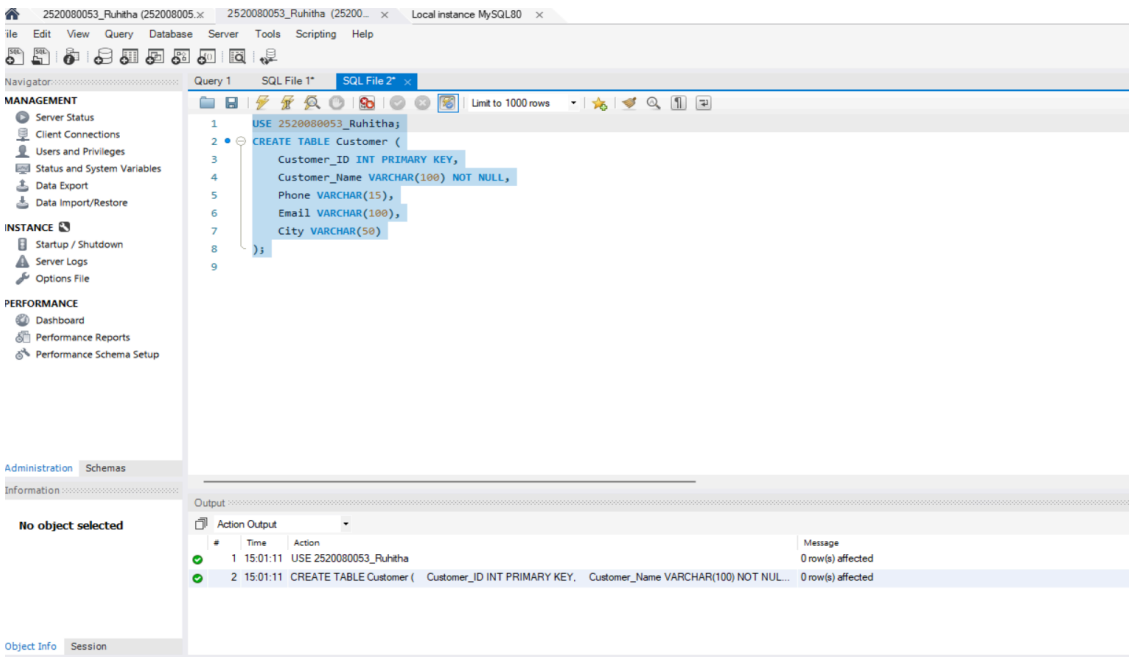
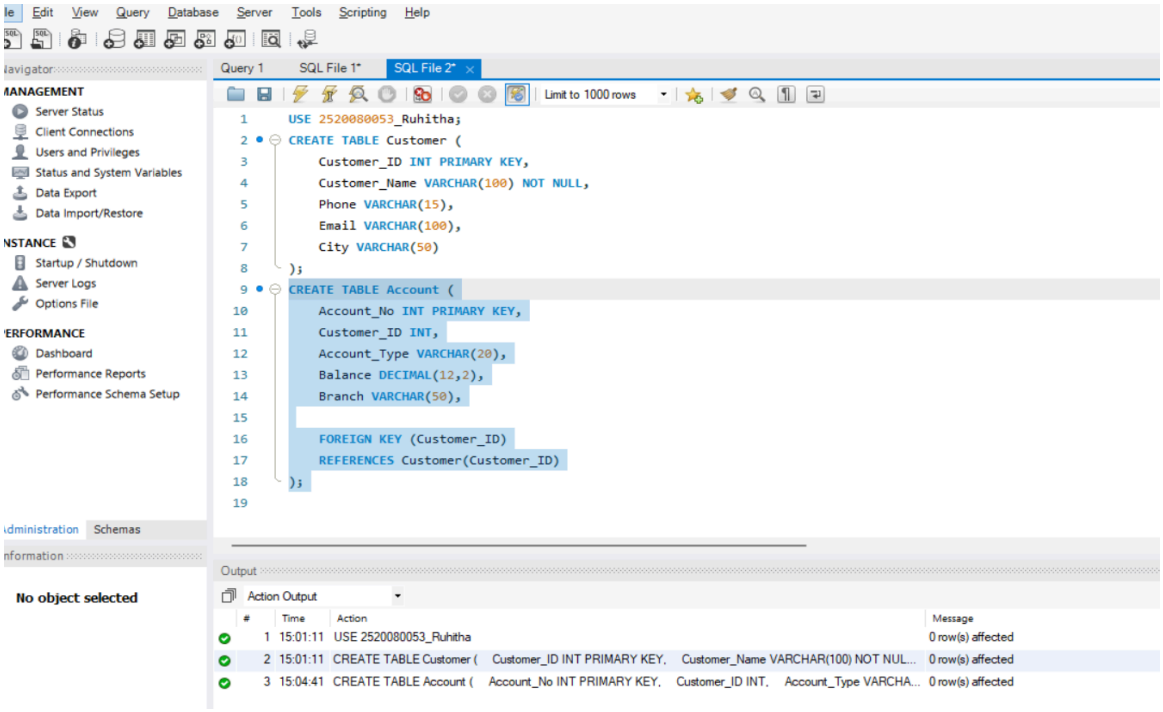
SKILL WEEK 3

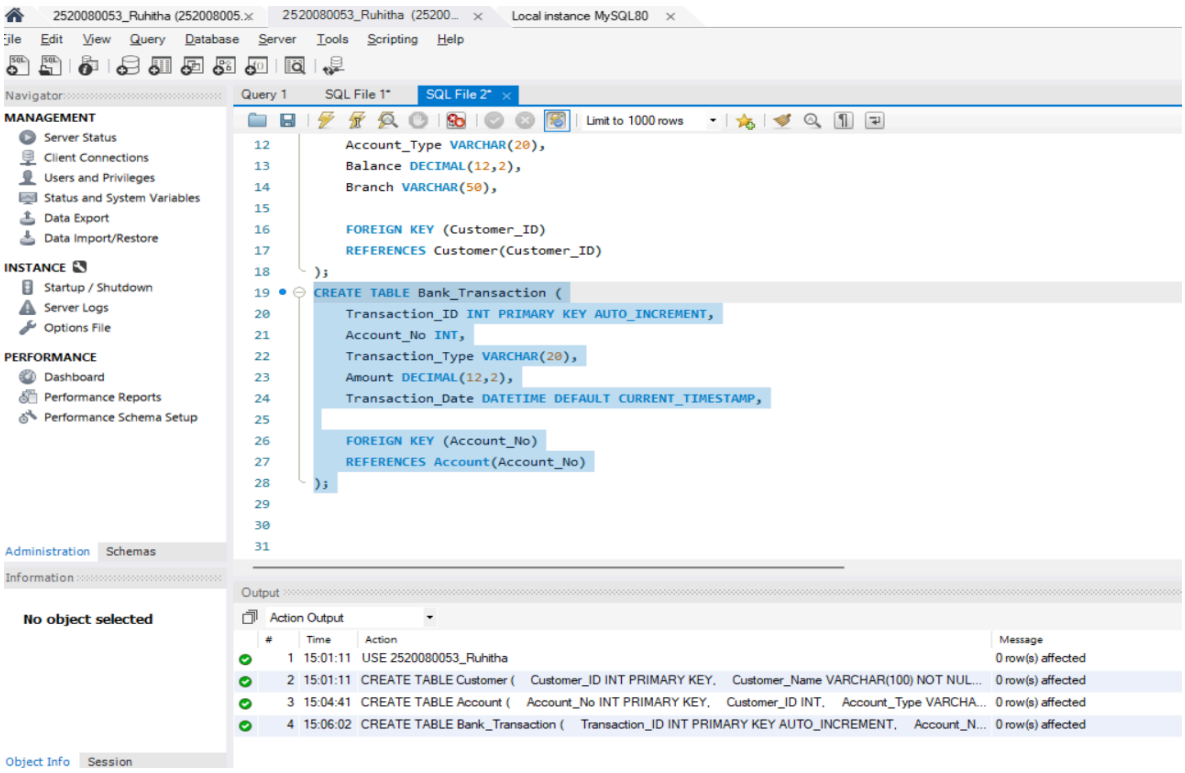
1) CREATE CUSTOMER TABLE



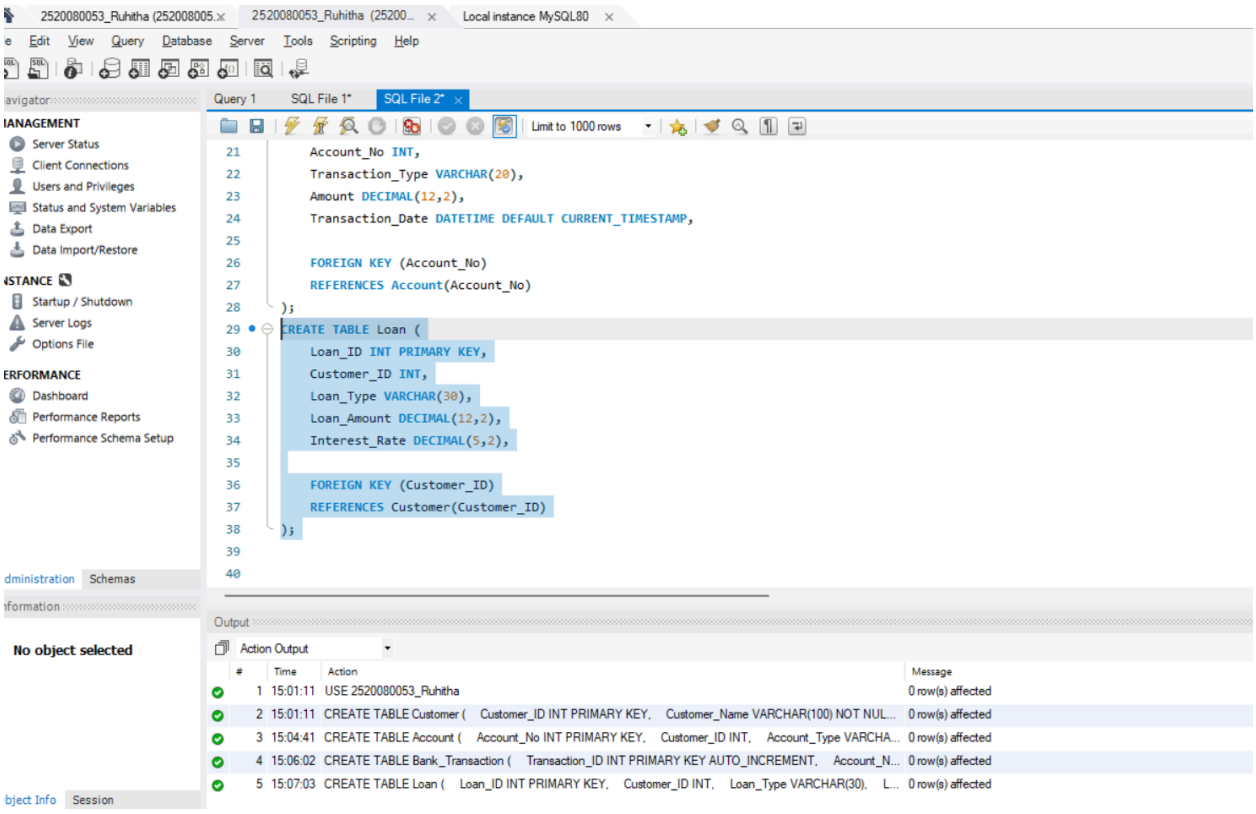
2) CREATE ACCOUNT TABLE



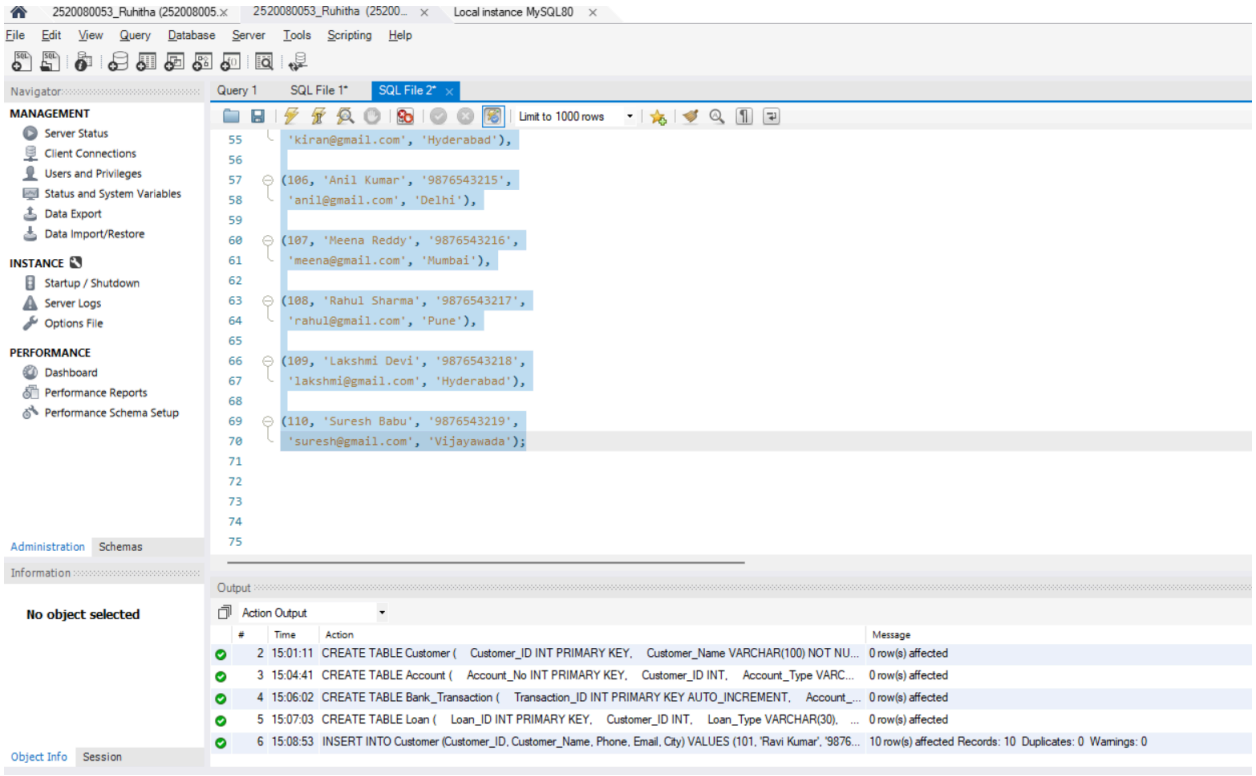
3) CREATE BANK TRANSACTION TABLE



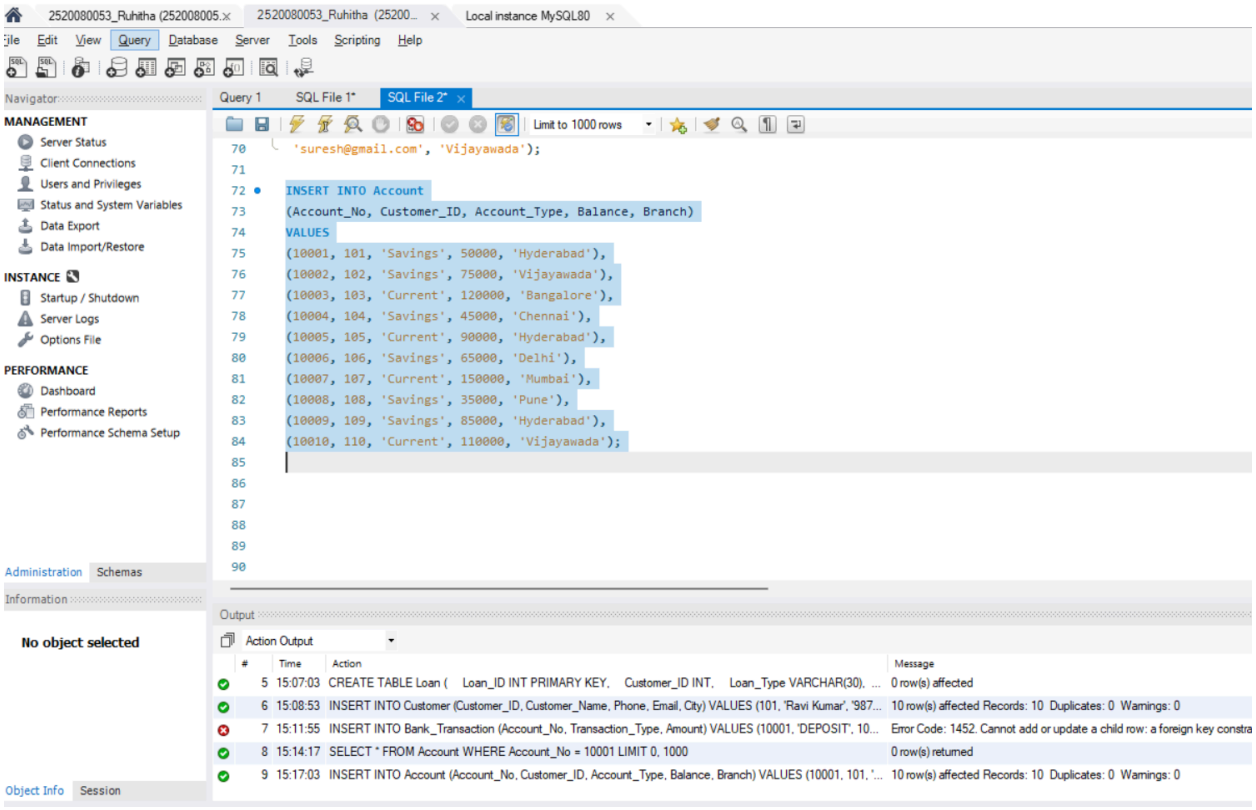
4) CREATE LOAN TABLE



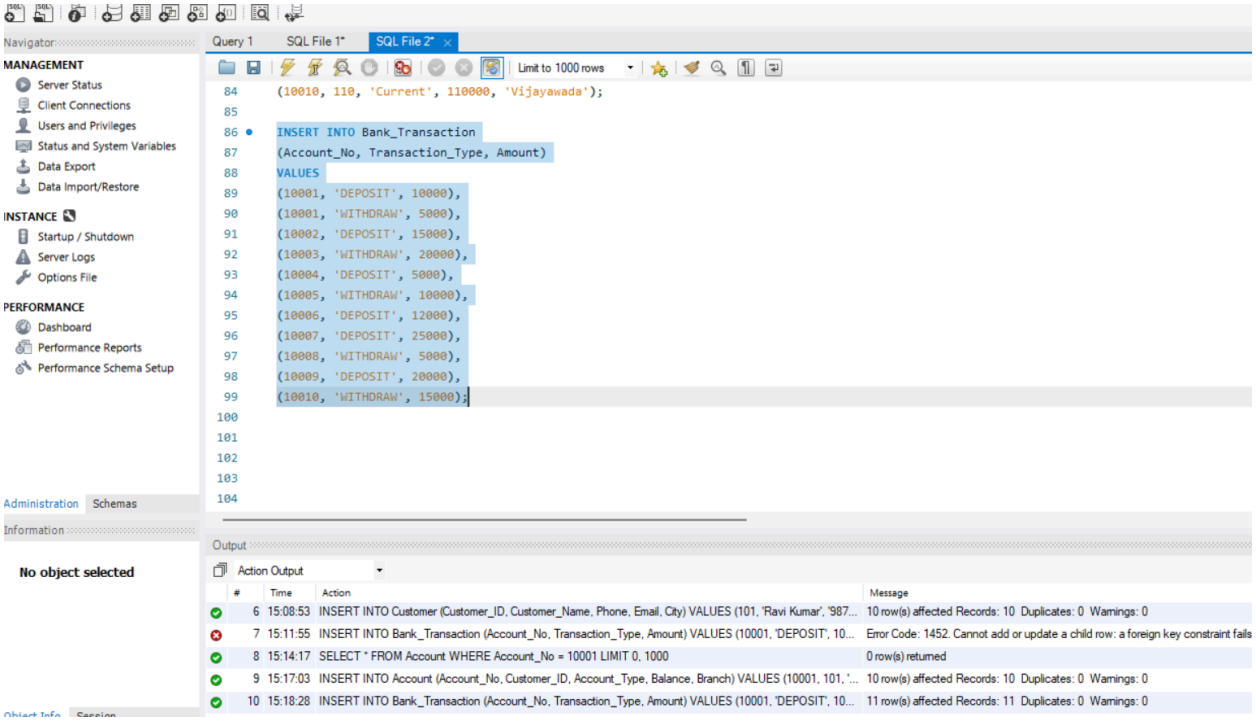
5) INSERT CUSTOMER DATA



6) INSERT ACCOUNT DATA



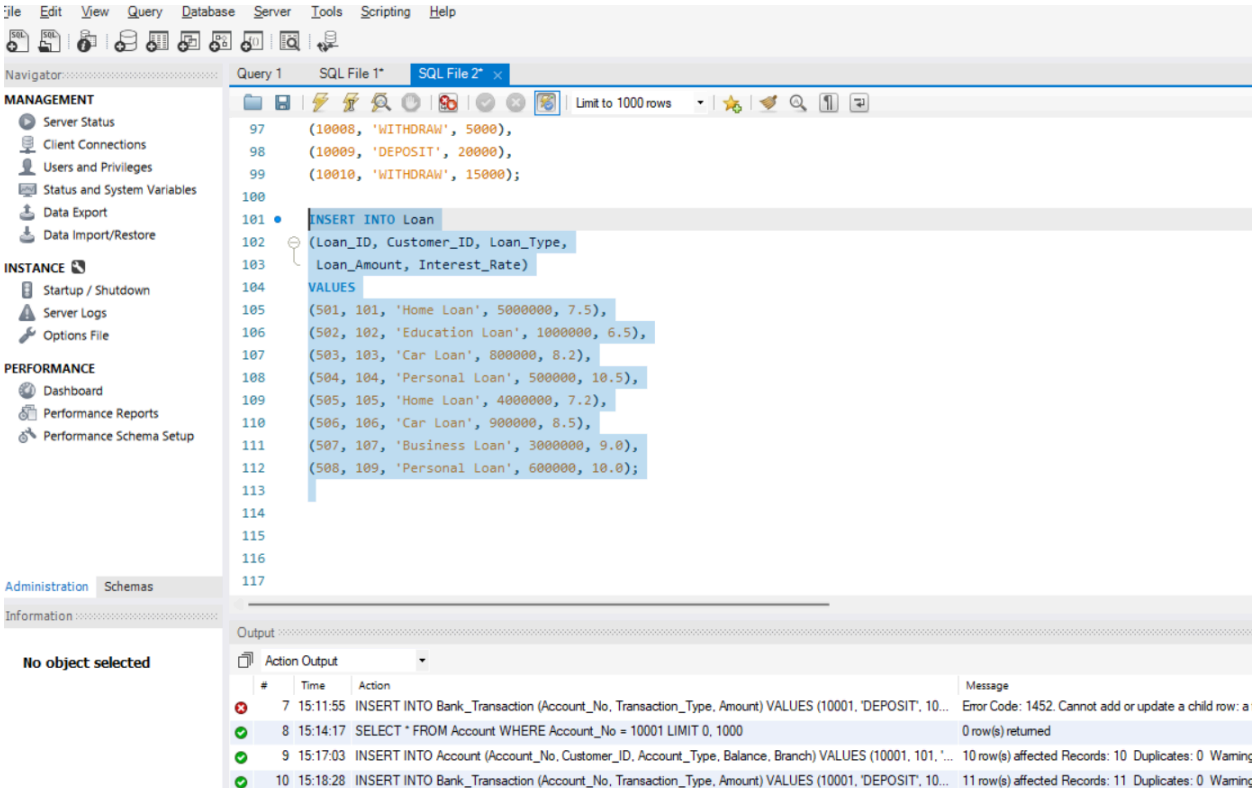
7)INSERT TRANSACTION DATA



```
(10010, 110, 'Current', 110000, 'Vijayawada');

INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10001, 'WITHDRAW', 5000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000),
(10006, 'DEPOSIT', 12000),
(10007, 'DEPOSIT', 25000),
(10008, 'WITHDRAW', 5000),
(10009, 'DEPOSIT', 20000),
(10010, 'WITHDRAW', 15000);
```

#	Time	Action	Message
6	15:08:53	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi Kumar', '987...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
7	15:11:55	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails
8	15:14:17	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	0 row(s) returned
9	15:17:03	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, '...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
10	15:18:28	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0

8)INSERT LOAN DATA

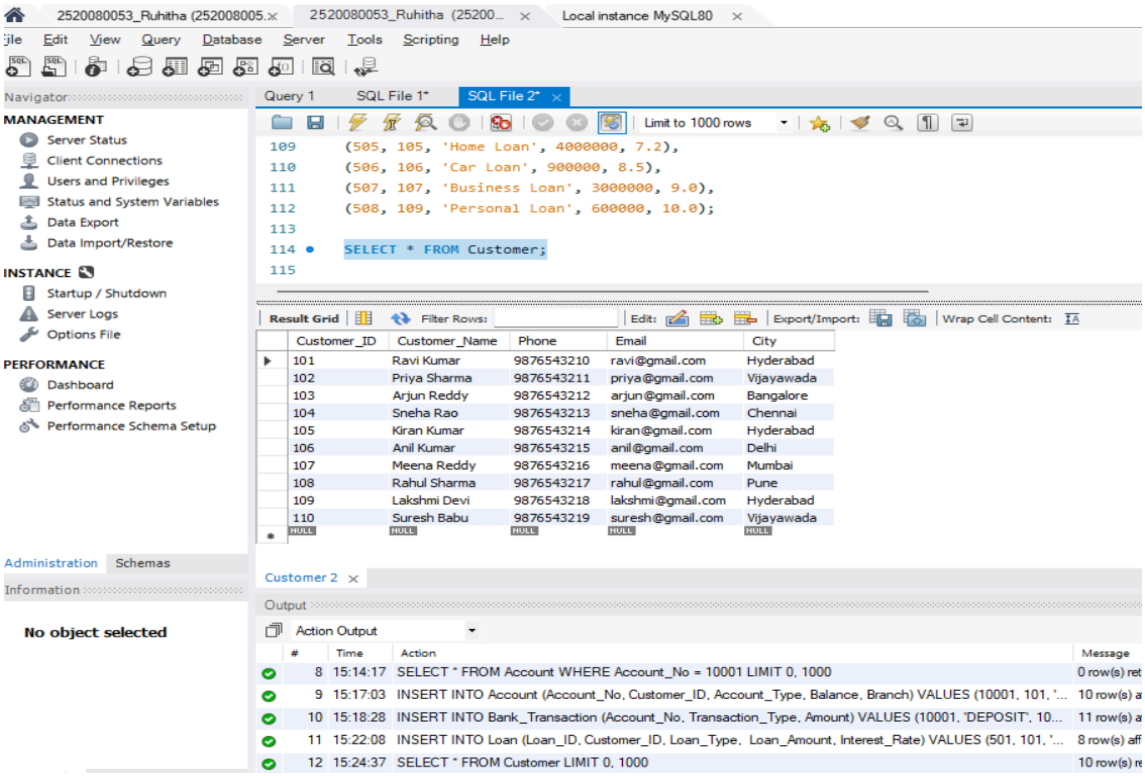


```
(10008, 'WITHDRAW', 5000),
(10009, 'DEPOSIT', 20000),
(10010, 'WITHDRAW', 15000);

INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type,
 Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),
(502, 102, 'Education Loan', 1000000, 6.5),
(503, 103, 'Car Loan', 800000, 8.2),
(504, 104, 'Personal Loan', 500000, 10.5),
(505, 105, 'Home Loan', 4000000, 7.2),
(506, 106, 'Car Loan', 900000, 8.5),
(507, 107, 'Business Loan', 3000000, 9.0),
(508, 109, 'Personal Loan', 600000, 10.0);
```

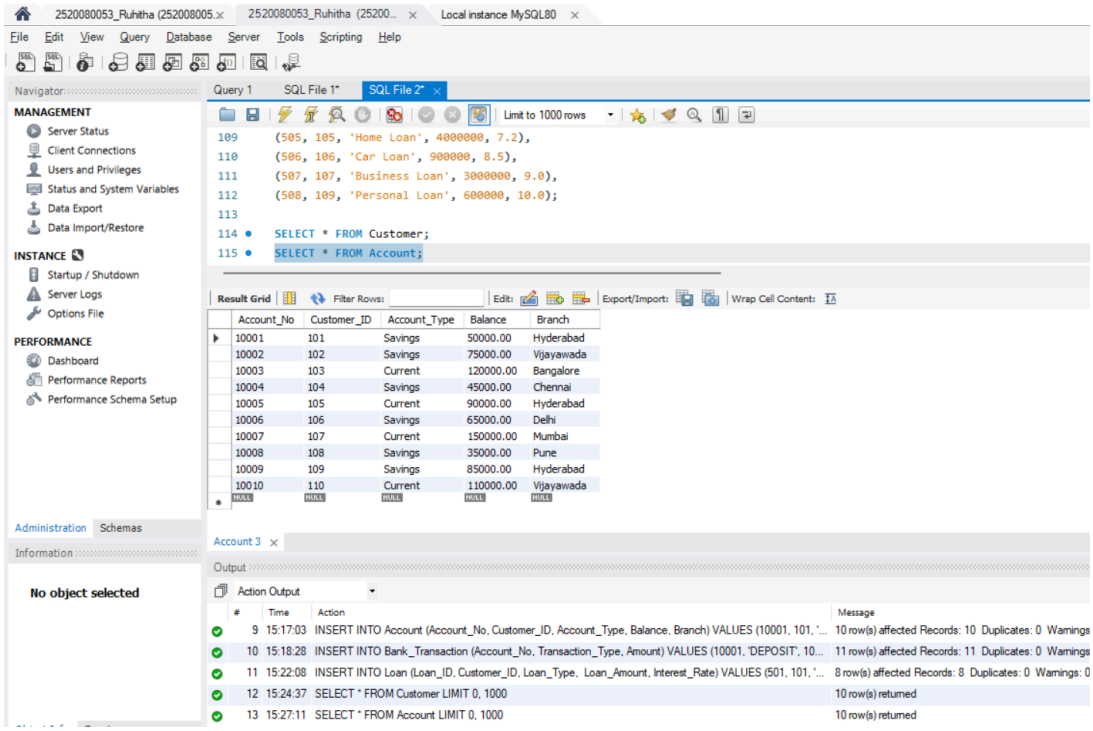
#	Time	Action	Message
7	15:11:55	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	Error Code: 1452. Cannot add or update a child row: a
8	15:14:17	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	0 row(s) returned
9	15:17:03	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, '...	10 row(s) affected Records: 10 Duplicates: 0 Warning
10	15:18:28	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	11 row(s) affected Records: 11 Duplicates: 0 Warning

9)DISPLAY ORIGINAL TABLES

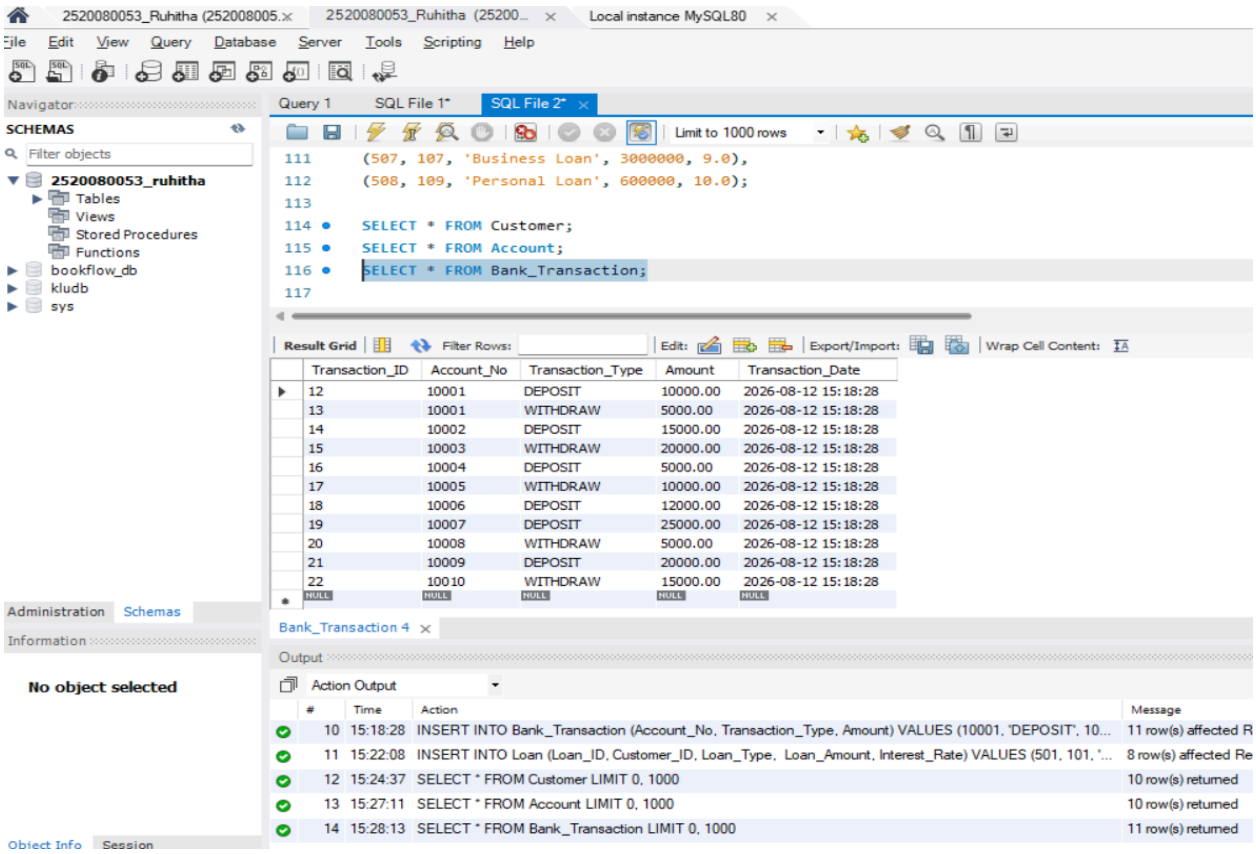
Customer



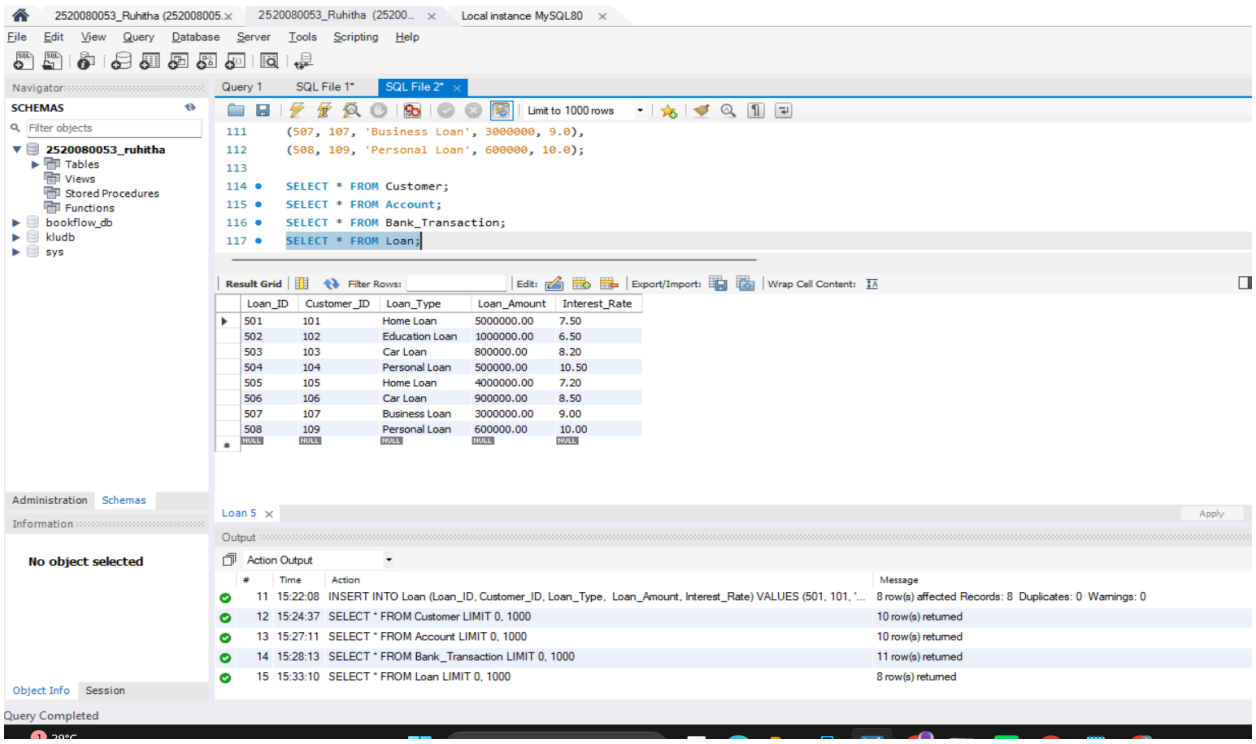
Account



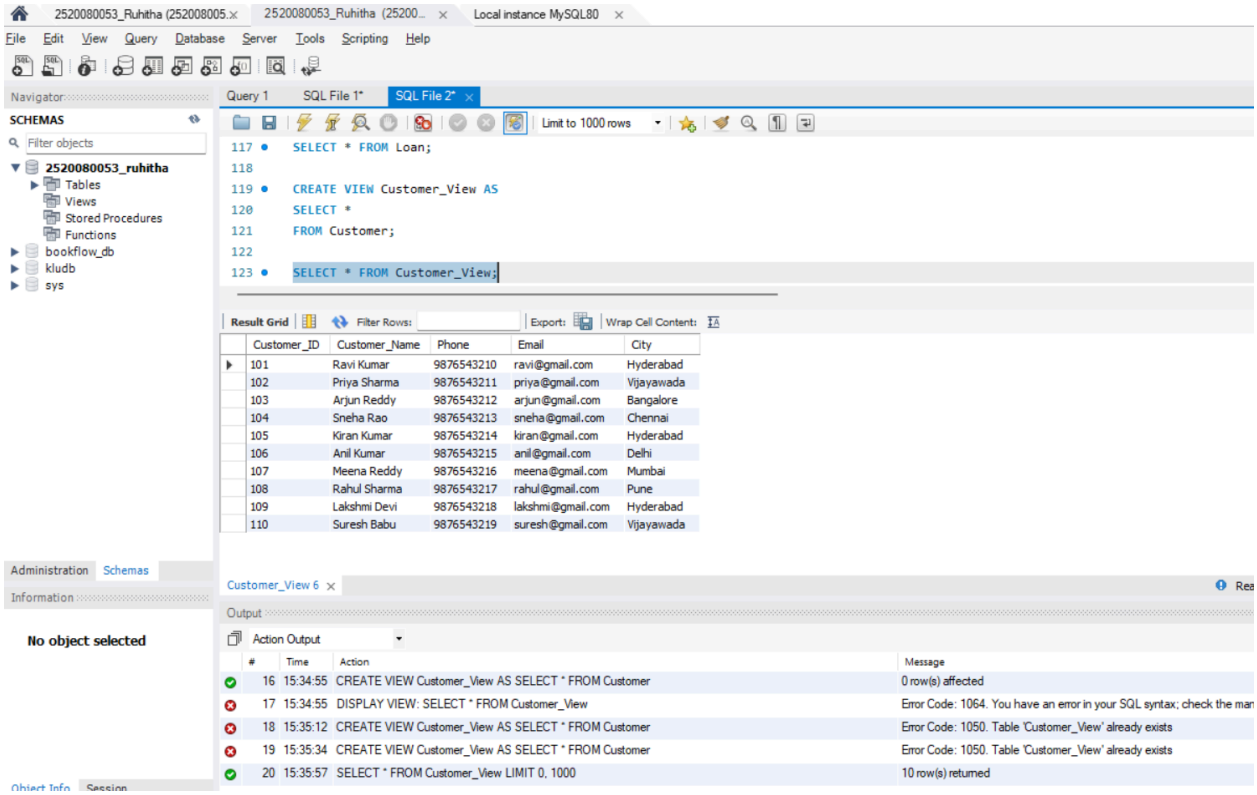
Bank_Transaction



Loan



CREATE A VIEW FOR ALL CUSTOMERS

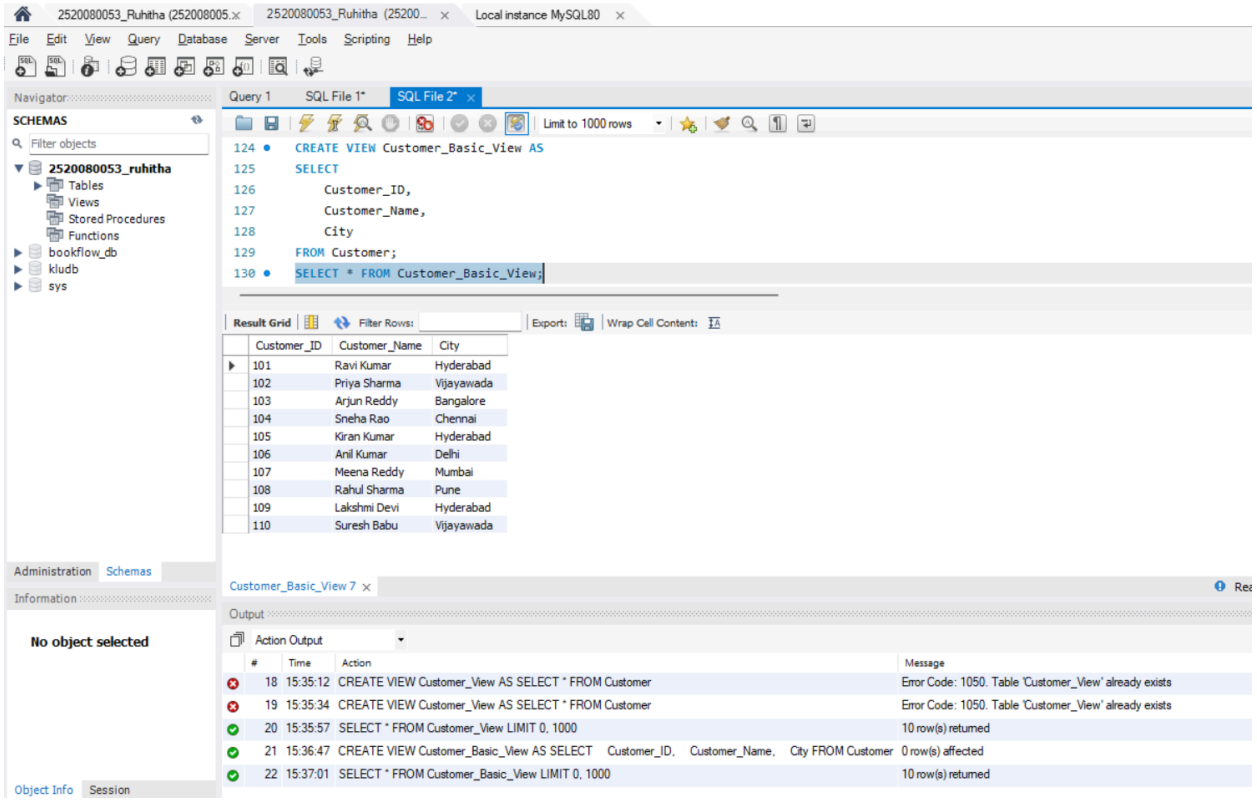


```
SELECT * FROM Loan;

CREATE VIEW Customer_View AS
SELECT *
FROM Customer;

SELECT * FROM Customer_View;
```

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune
109	Lakshmi Devi	9876543218	lakshmi@gmail.com	Hyderabad
110	Suresh Babu	9876543219	suresh@gmail.com	Vijayawada

#	Time	Action	Message
16	15:34:55	CREATE VIEW Customer_View AS SELECT * FROM Customer	0 row(s) affected
17	15:34:55	DISPLAY VIEW: SELECT * FROM Customer_View	Error Code: 1064. You have an error in your SQL syntax; check the man
18	15:35:12	CREATE VIEW Customer_View AS SELECT * FROM Customer	Error Code: 1050. Table 'Customer_View' already exists
19	15:35:34	CREATE VIEW Customer_View AS SELECT * FROM Customer	Error Code: 1050. Table 'Customer_View' already exists
20	15:35:57	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned

CREATE A VIEW FOR CUSTOMER BASIC INFORMATION

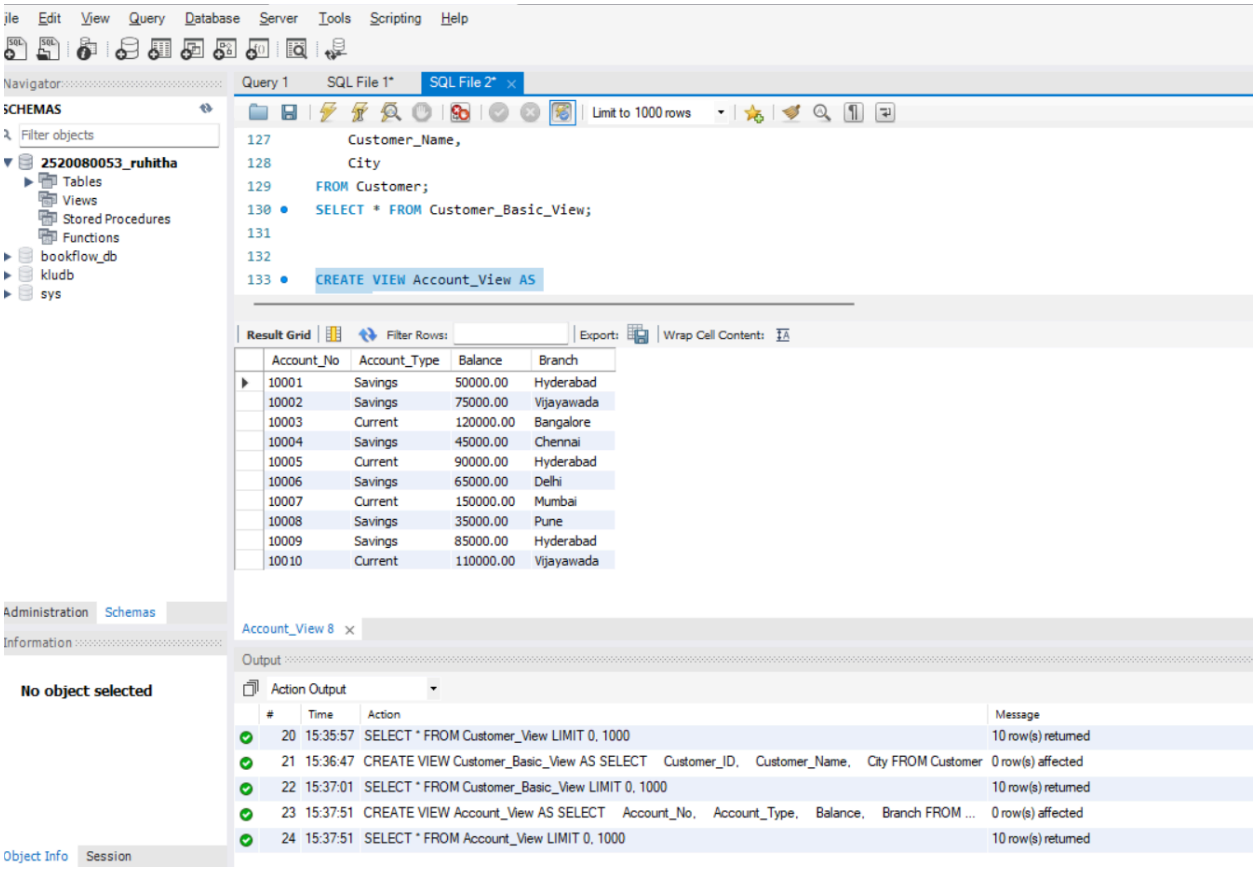


```
CREATE VIEW Customer_Basic_View AS
SELECT
    Customer_ID,
    Customer_Name,
    City
FROM Customer;
SELECT * FROM Customer_Basic_View;
```

Customer_ID	Customer_Name	City
101	Ravi Kumar	Hyderabad
102	Priya Sharma	Vijayawada
103	Arjun Reddy	Bangalore
104	Sneha Rao	Chennai
105	Kiran Kumar	Hyderabad
106	Anil Kumar	Delhi
107	Meena Reddy	Mumbai
108	Rahul Sharma	Pune
109	Lakshmi Devi	Hyderabad
110	Suresh Babu	Vijayawada

#	Time	Action	Message
18	15:35:12	CREATE VIEW Customer_View AS SELECT * FROM Customer	Error Code: 1050. Table 'Customer_View' already exists
19	15:35:34	CREATE VIEW Customer_View AS SELECT * FROM Customer	Error Code: 1050. Table 'Customer_View' already exists
20	15:35:57	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
21	15:36:47	CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer	0 row(s) affected
22	15:37:01	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned

CREATE A VIEW FOR ACCOUNT INFORMATION

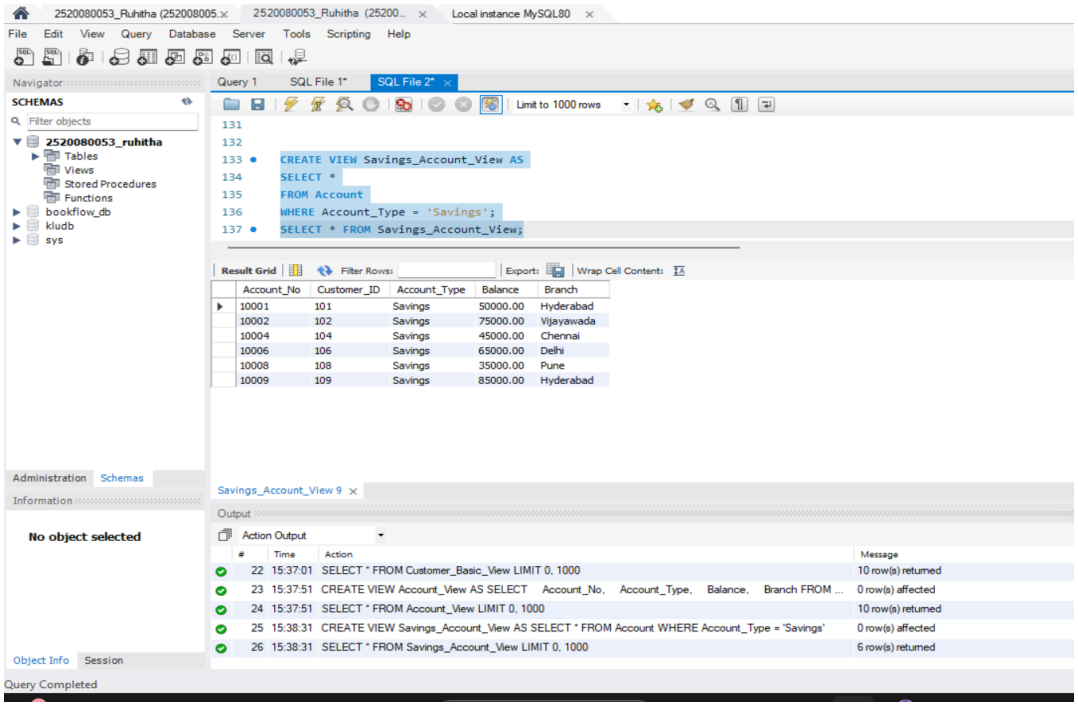


```
Customer_Name,
City
FROM Customer;
SELECT * FROM Customer_Basic_View;


CREATE VIEW Account_View AS
```

Account_No	Account_Type	Balance	Branch
10001	Savings	50000.00	Hyderabad
10002	Savings	75000.00	Vijayawada
10003	Current	120000.00	Bangalore
10004	Savings	45000.00	Chennai
10005	Current	90000.00	Hyderabad
10006	Savings	65000.00	Delhi
10007	Current	150000.00	Mumbai
10008	Savings	35000.00	Pune
10009	Savings	85000.00	Hyderabad
10010	Current	110000.00	Vijayawada

#	Time	Action	Message
20	15:35:57	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
21	15:36:47	CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer	0 row(s) affected
22	15:37:01	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned
23	15:37:51	CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM ...	0 row(s) affected
24	15:37:51	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned

CREATE A VIEW FOR SAVINGS ACCOUNTS

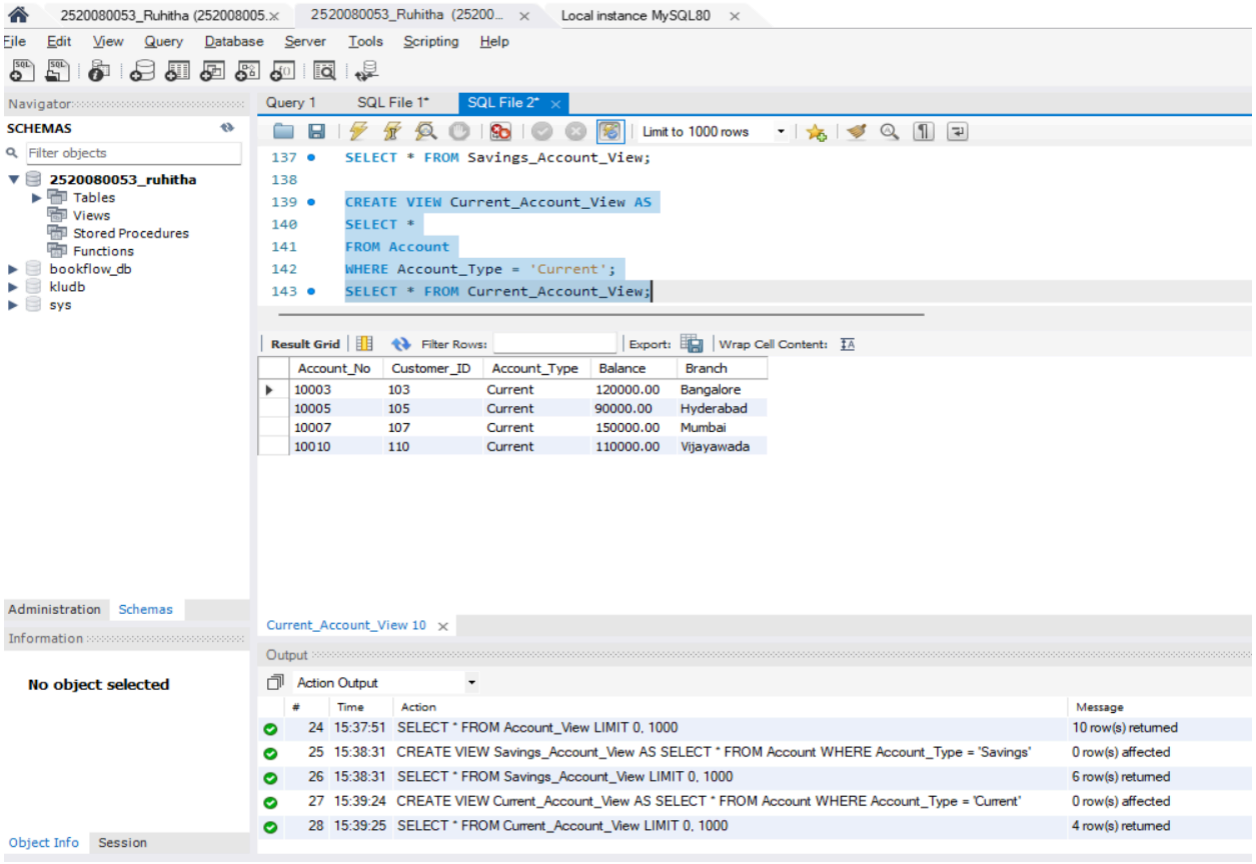


```
CREATE VIEW Savings_Account_View AS
SELECT *
FROM Account
WHERE Account_Type = 'Savings';
SELECT * FROM Savings_Account_View;
```

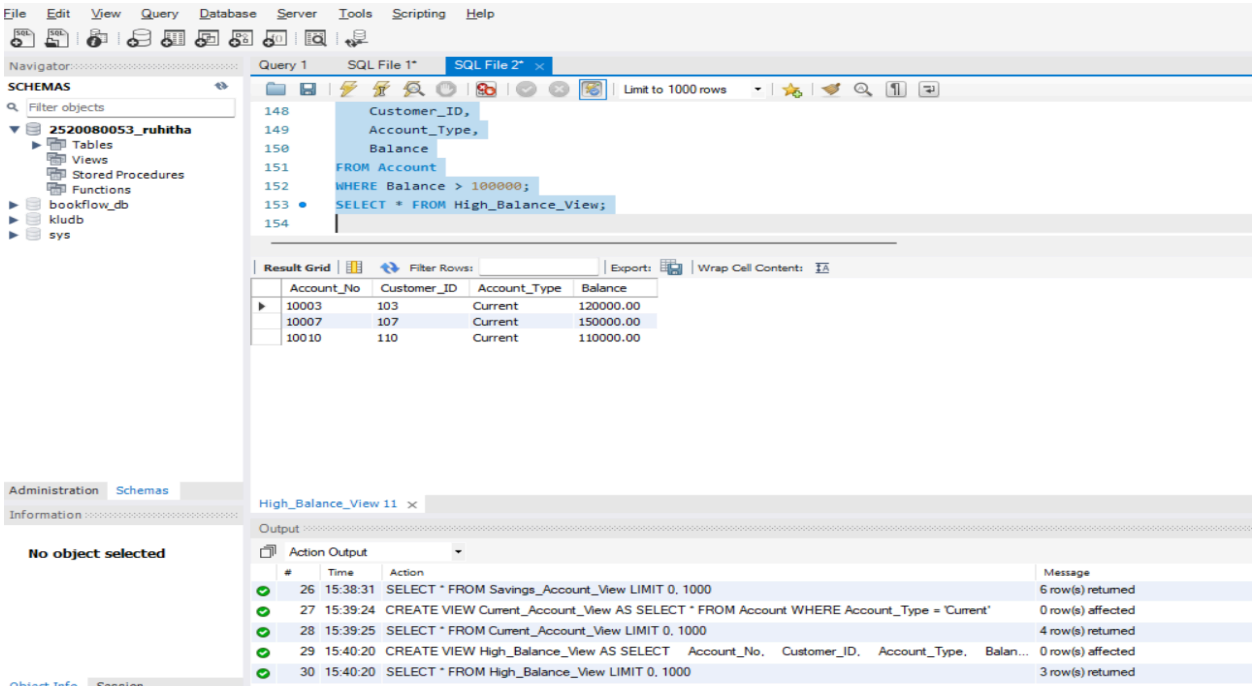
Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10004	104	Savings	45000.00	Chennai
10006	106	Savings	65000.00	Delhi
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad

#	Time	Action	Message
22	15:37:01	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned
23	15:37:51	CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM ...	0 row(s) affected
24	15:37:51	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned
25	15:38:31	CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Savings'	0 row(s) affected
26	15:38:31	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 row(s) returned

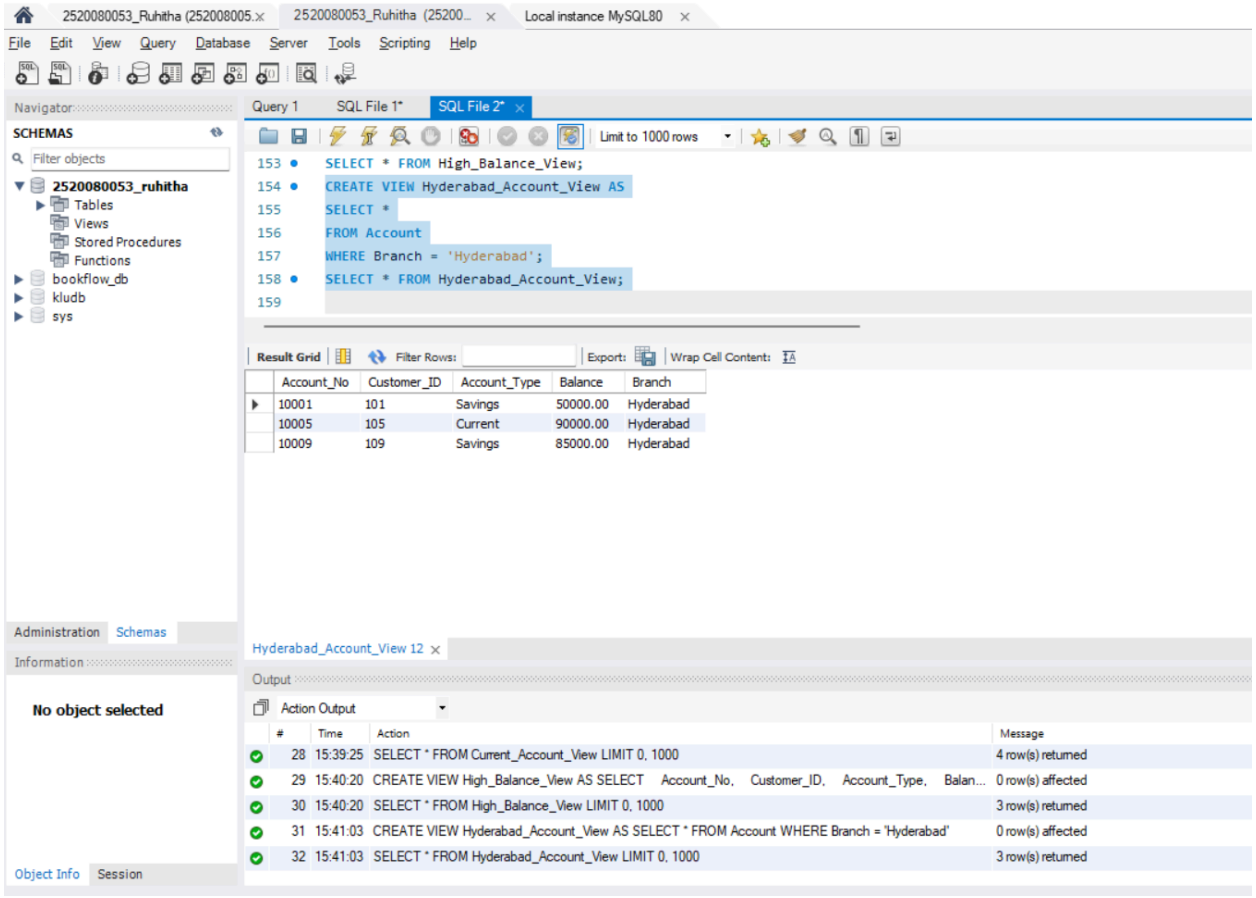
CREATE A VIEW FOR CURRENT ACCOUNTS



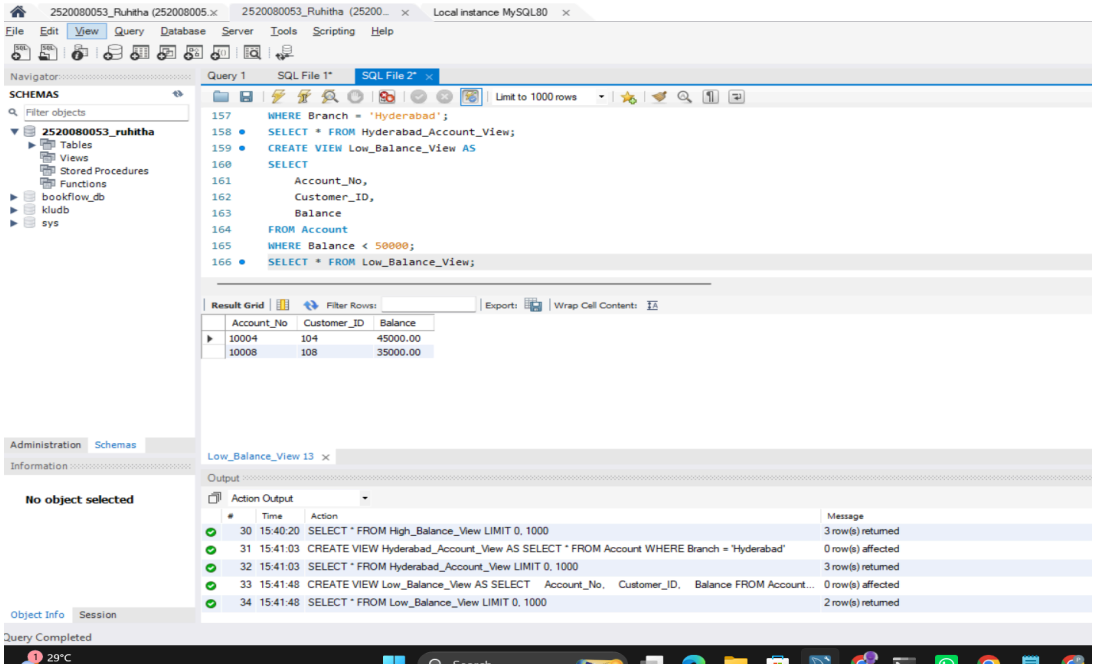
HIGH BALANCE ACCOUNT VIEW



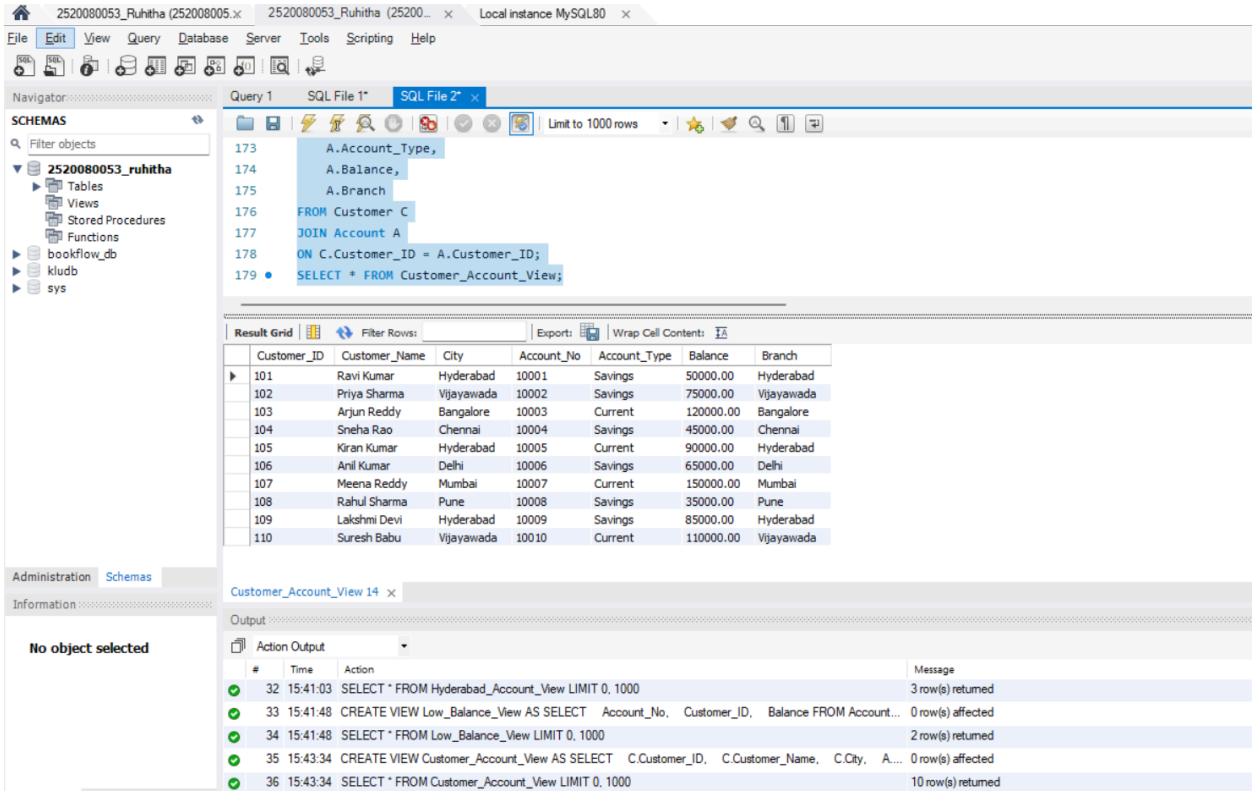
HYDERABAD ACCOUNT VIEW



LOW BALANCE ACCOUNT VIEW



CUSTOMER + ACCOUNT VIEW

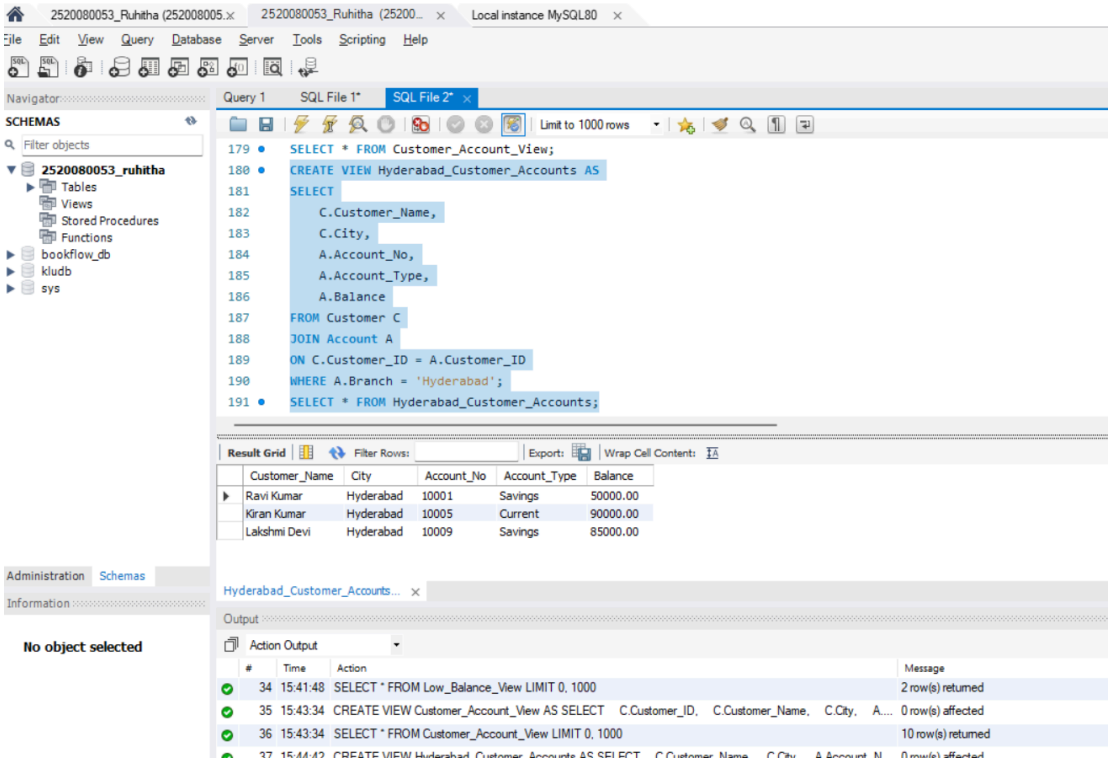


```
    A.Account_Type,
    A.Balance,
    A.Branch
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID;
SELECT * FROM Customer_Account_View;
```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	Hyderabad	10001	Savings	50000.00	Hyderabad
102	Priya Sharma	Vijayawada	10002	Savings	75000.00	Vijayawada
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad
106	Anil Kumar	Delhi	10006	Savings	65000.00	Delhi
107	Meena Reddy	Mumbai	10007	Current	150000.00	Mumbai
108	Rahul Sharma	Pune	10008	Savings	35000.00	Pune
109	Lakshmi Devi	Hyderabad	10009	Savings	85000.00	Hyderabad
110	Suresh Babu	Vijayawada	10010	Current	110000.00	Vijayawada

#	Time	Action	Message
32	15:41:03	SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000	3 row(s) returned
33	15:41:48	CREATE VIEW Low_Balance_View AS SELECT Account_No, Customer_ID, Balance FROM Account...	0 row(s) affected
34	15:41:48	SELECT * FROM Low_Balance_View LIMIT 0, 1000	2 row(s) returned
35	15:43:34	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A....	0 row(s) affected
36	15:43:34	SELECT * FROM Customer_Account_View LIMIT 0, 1000	10 row(s) returned

HYDERABAD CUSTOMER ACCOUNT VIEW

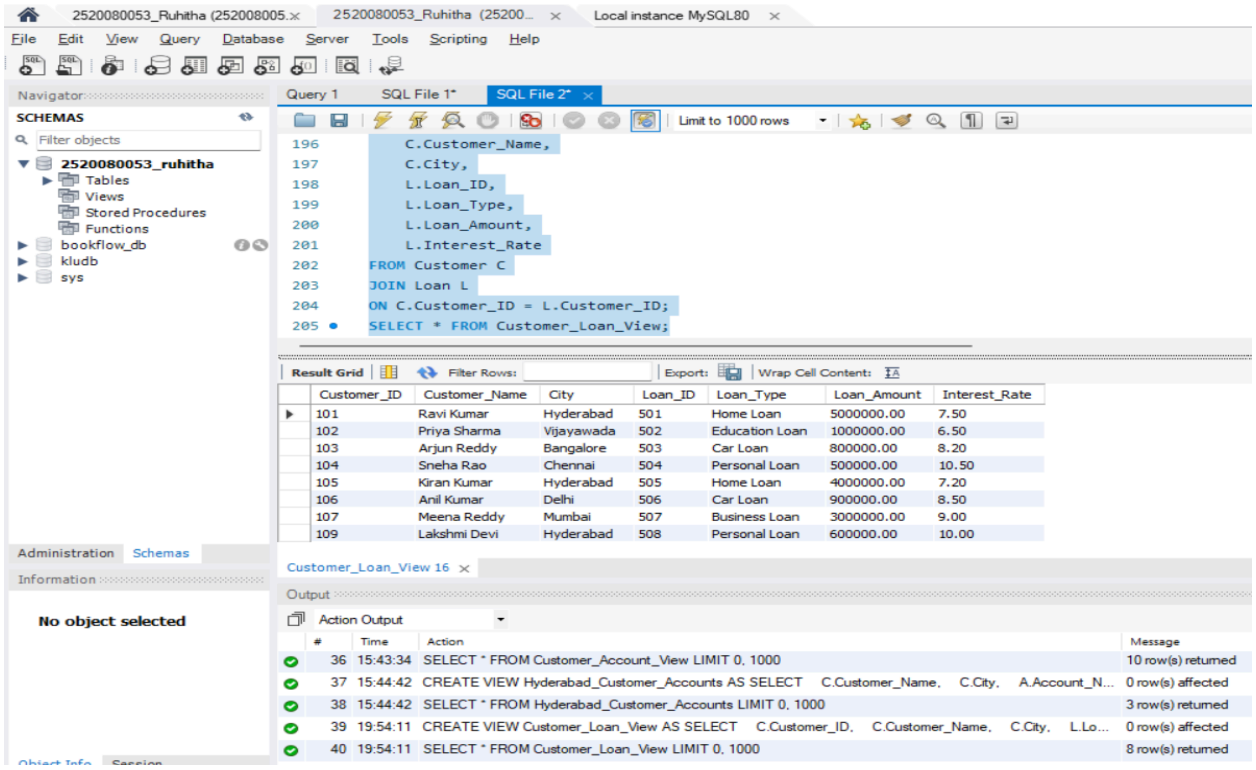


```
SELECT * FROM Customer_Account_View;
CREATE VIEW Hyderabad_Customer_Accounts AS
SELECT
    C.Customer_Name,
    C.City,
    A.Account_No,
    A.Account_Type,
    A.Balance
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID
WHERE A.Branch = 'Hyderabad';
SELECT * FROM Hyderabad_Customer_Accounts;
```

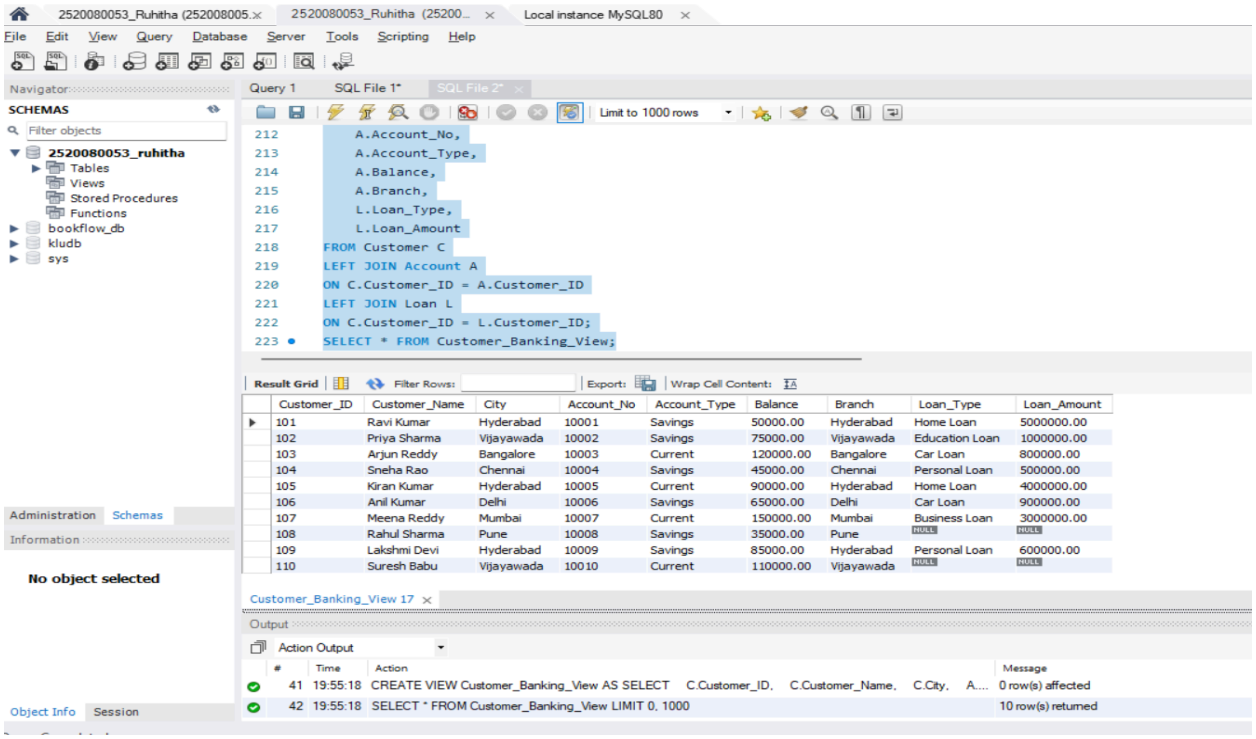
Customer_Name	City	Account_No	Account_Type	Balance
Ravi Kumar	Hyderabad	10001	Savings	50000.00
Kiran Kumar	Hyderabad	10005	Current	90000.00
Lakshmi Devi	Hyderabad	10009	Savings	85000.00

#	Time	Action	Message
34	15:41:48	SELECT * FROM Low_Balance_View LIMIT 0, 1000	2 row(s) returned
35	15:43:34	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A....	0 row(s) affected
36	15:43:34	SELECT * FROM Customer_Account_View LIMIT 0, 1000	10 row(s) returned

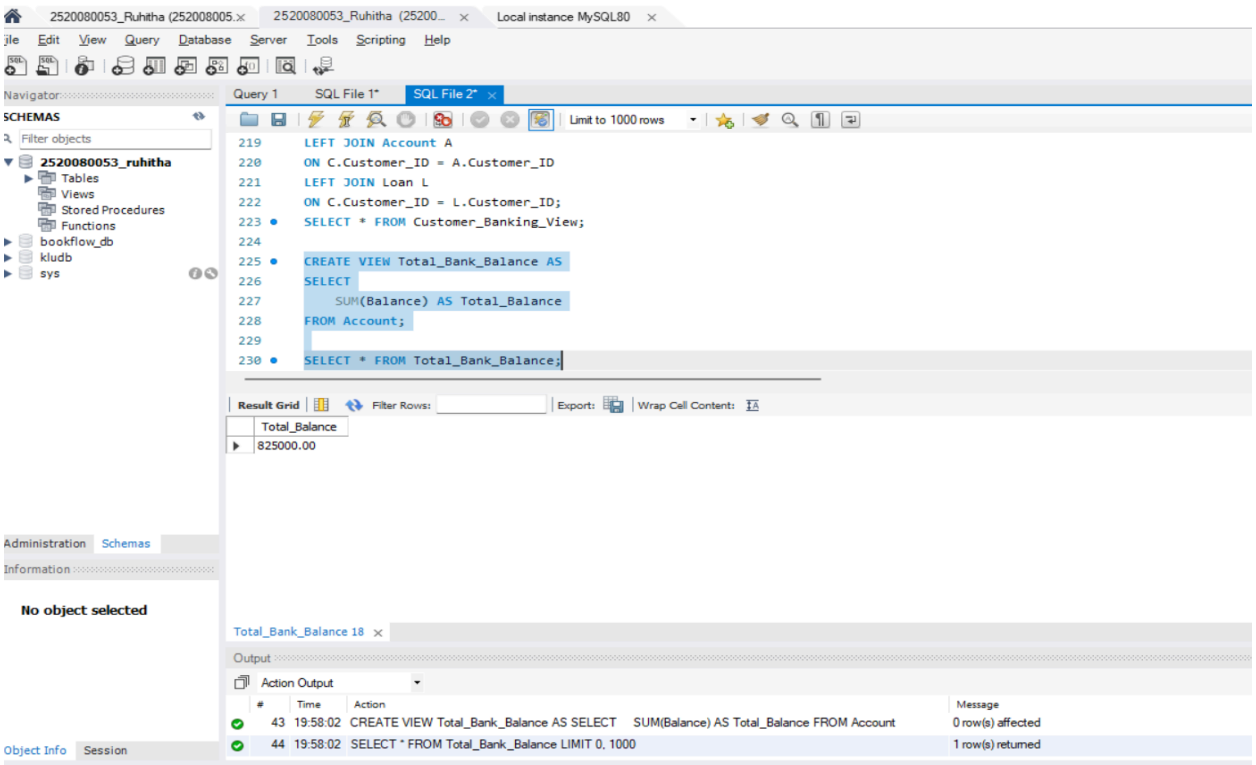
CUSTOMER LOAN VIEW



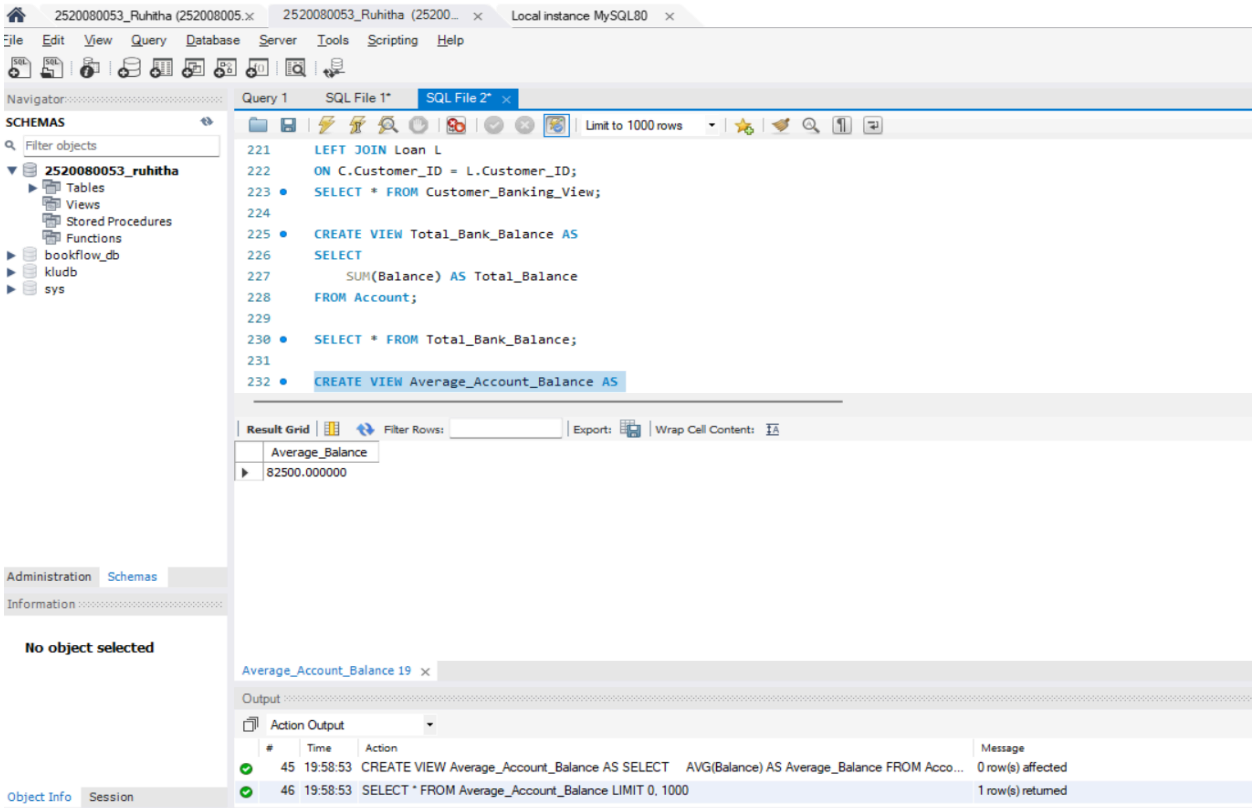
COMPLETE CUSTOMER BANKING VIEW



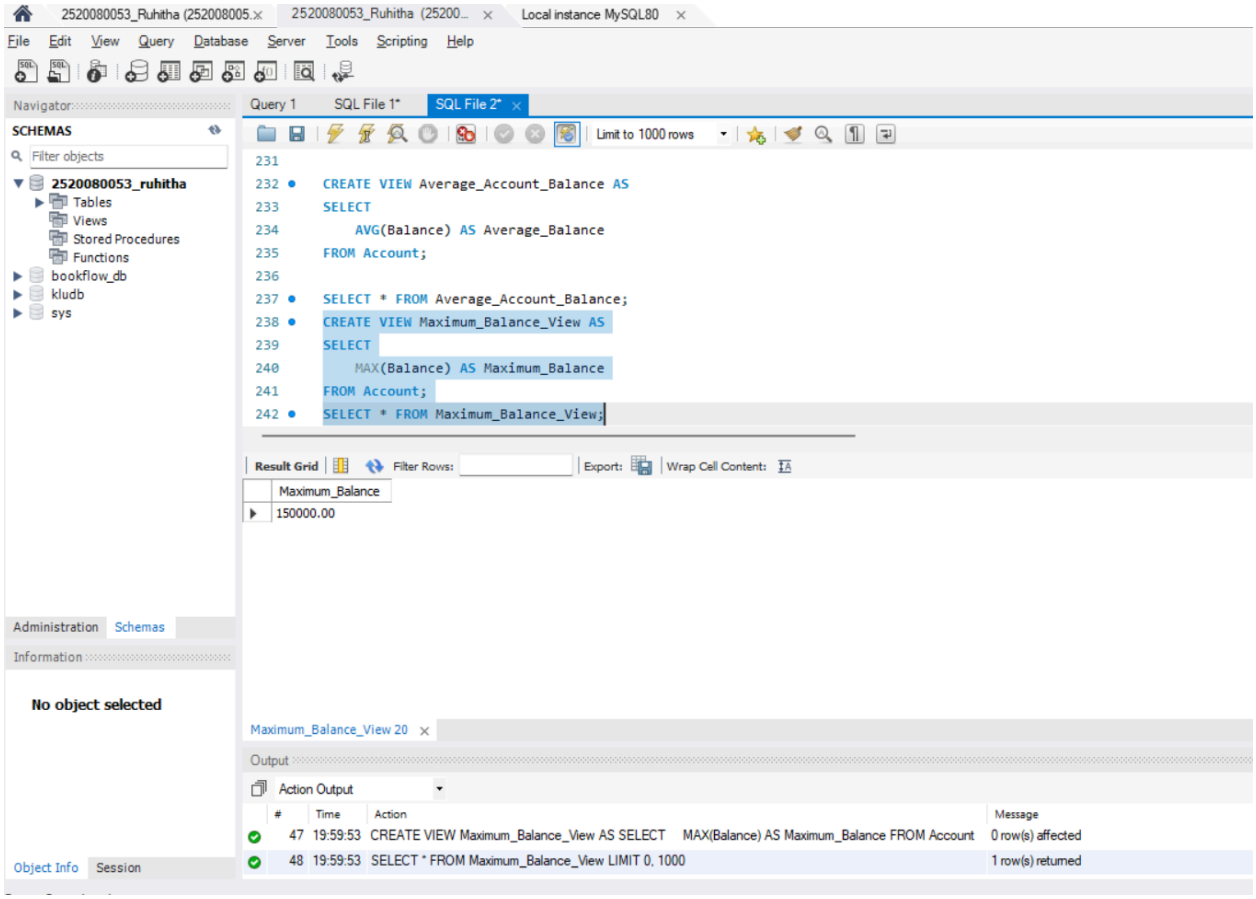
TOTAL BANK BALANCE



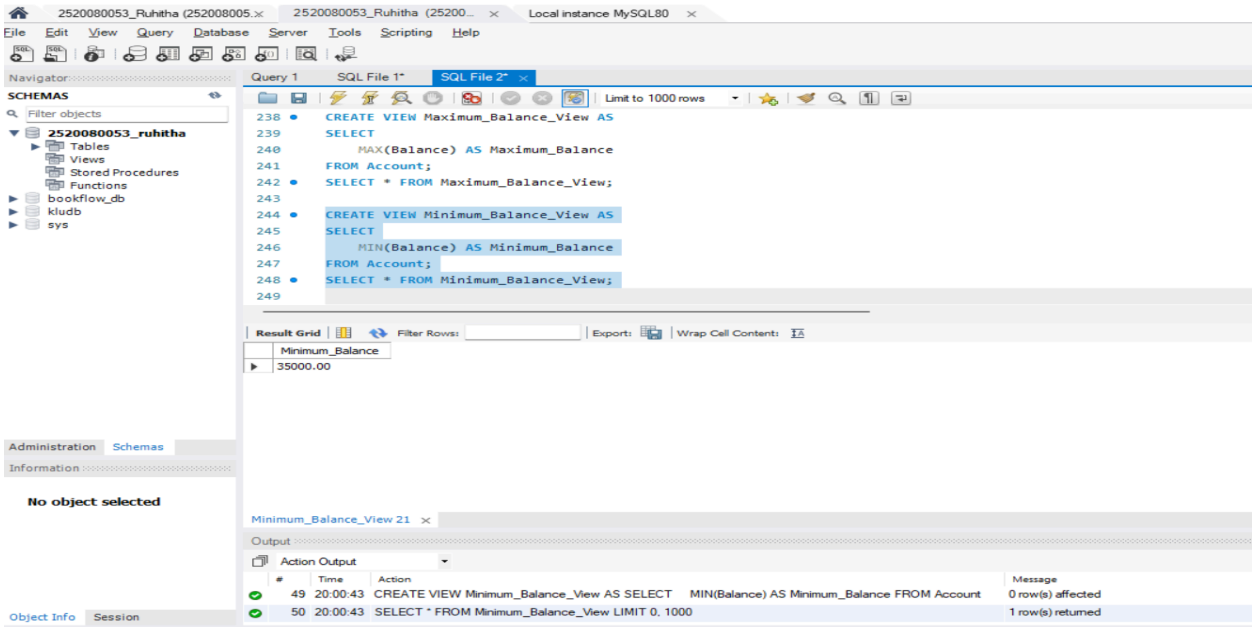
AVERAGE ACCOUNT BALANCE



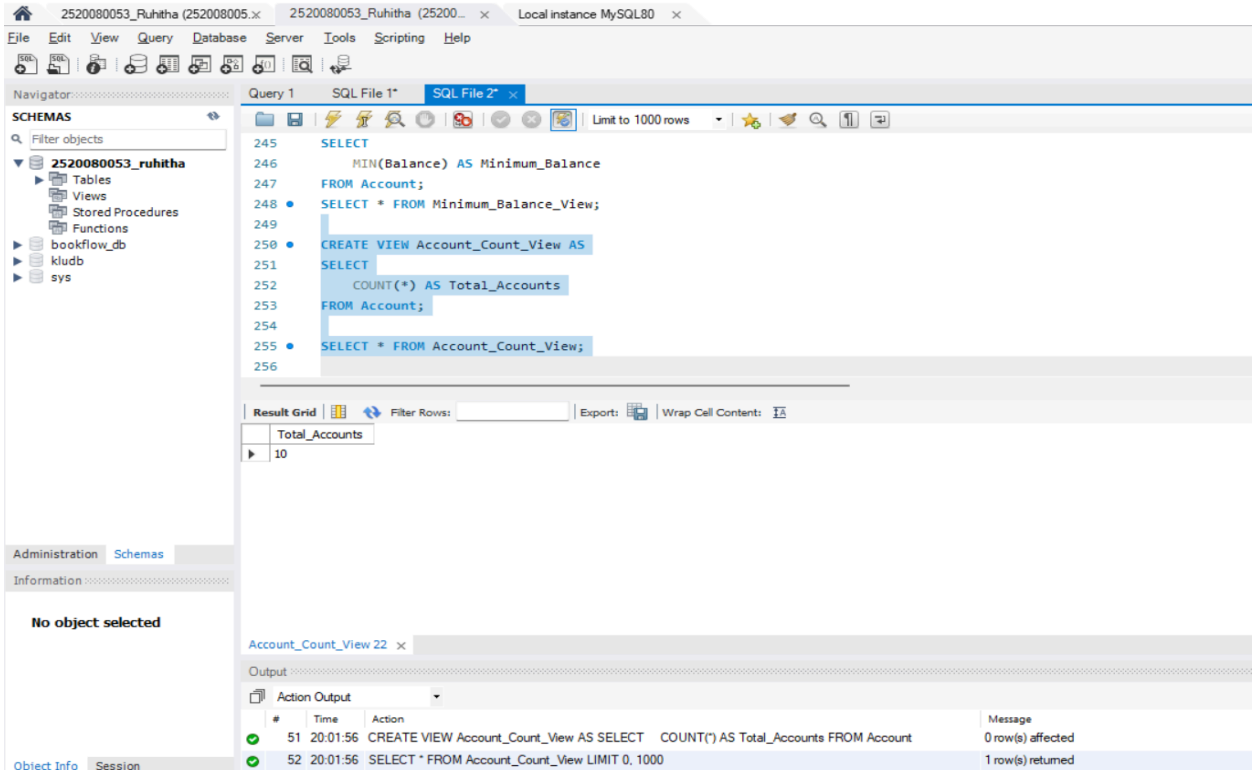
MAXIMUM ACCOUNT BALANCE



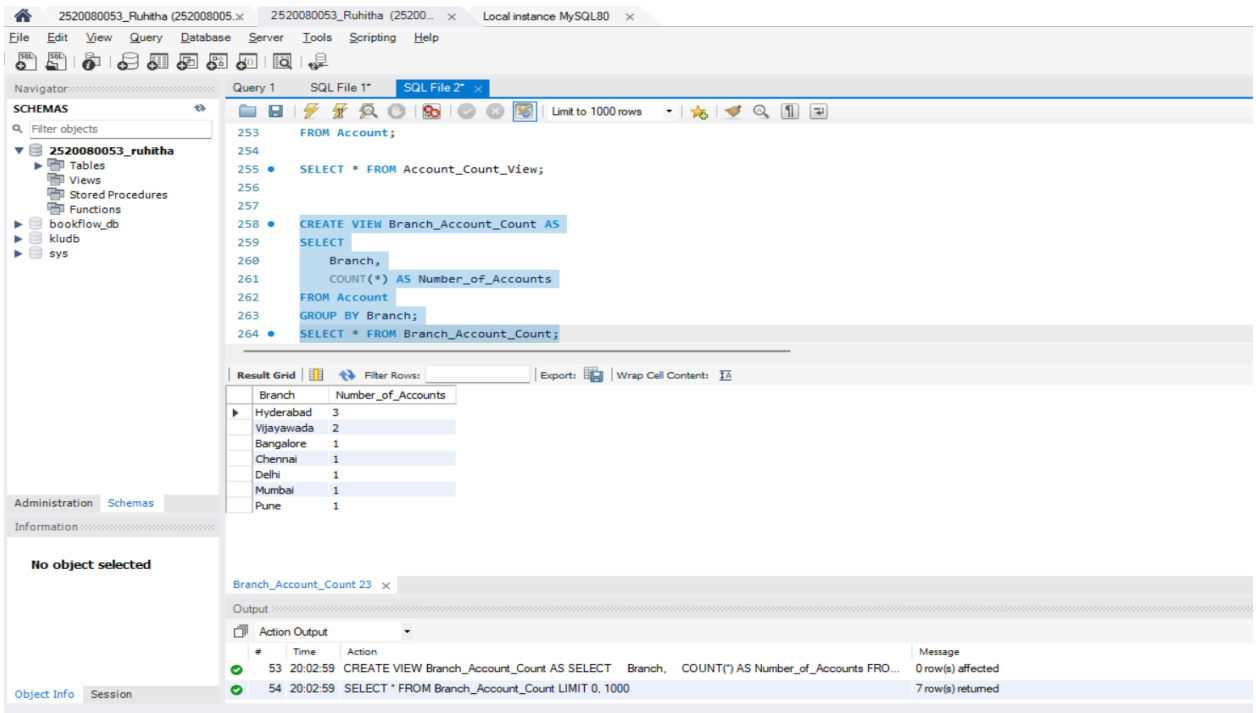
MINIMUM ACCOUNT BALANCE



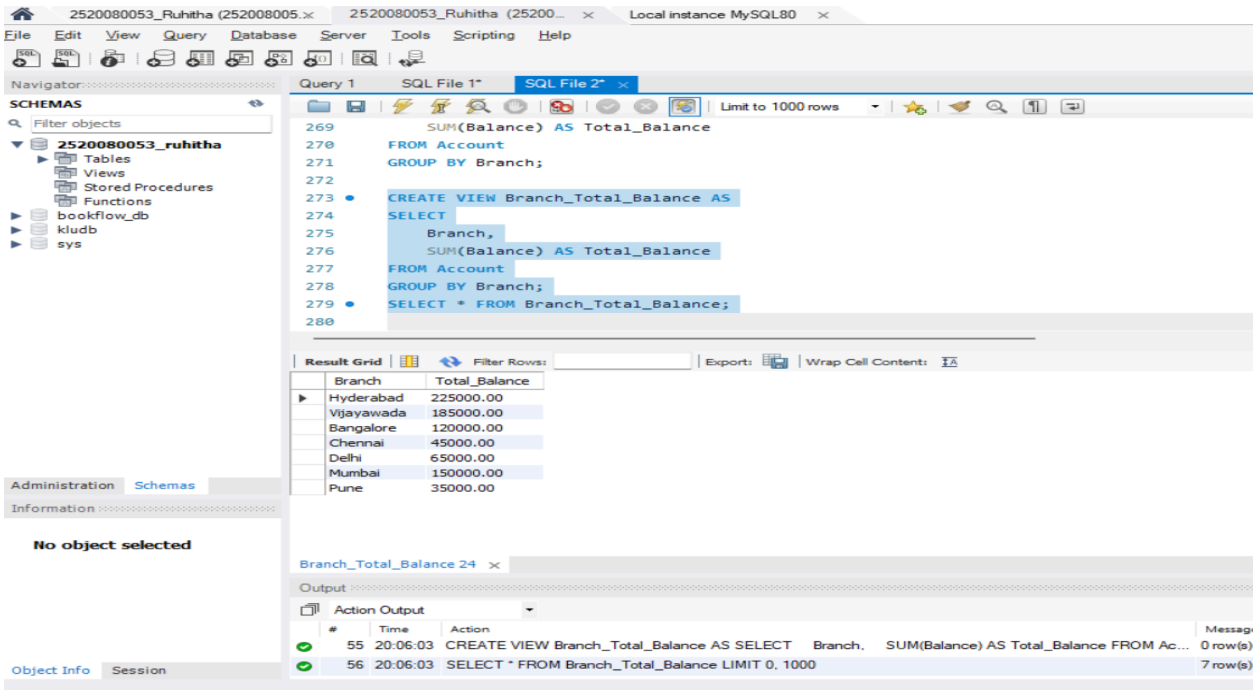
TOTAL NUMBER OF ACCOUNTS



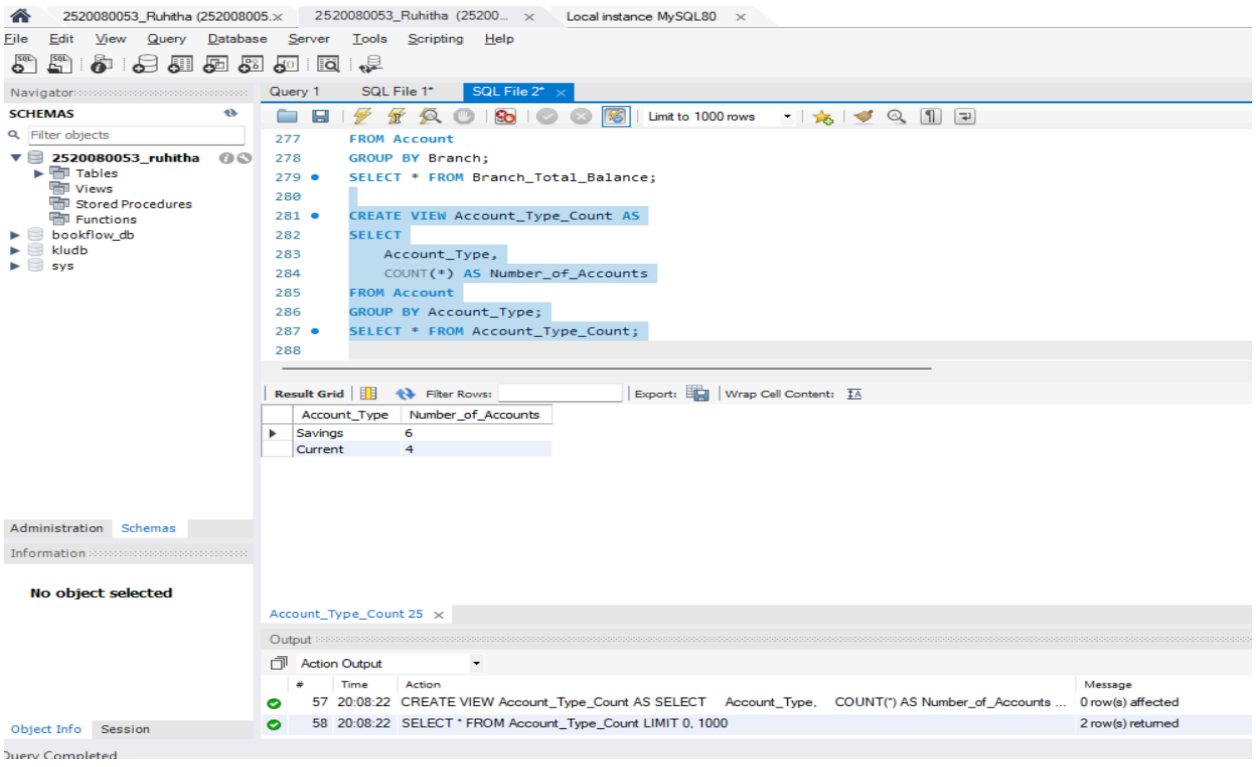
BRANCH-WISE ACCOUNT COUNT



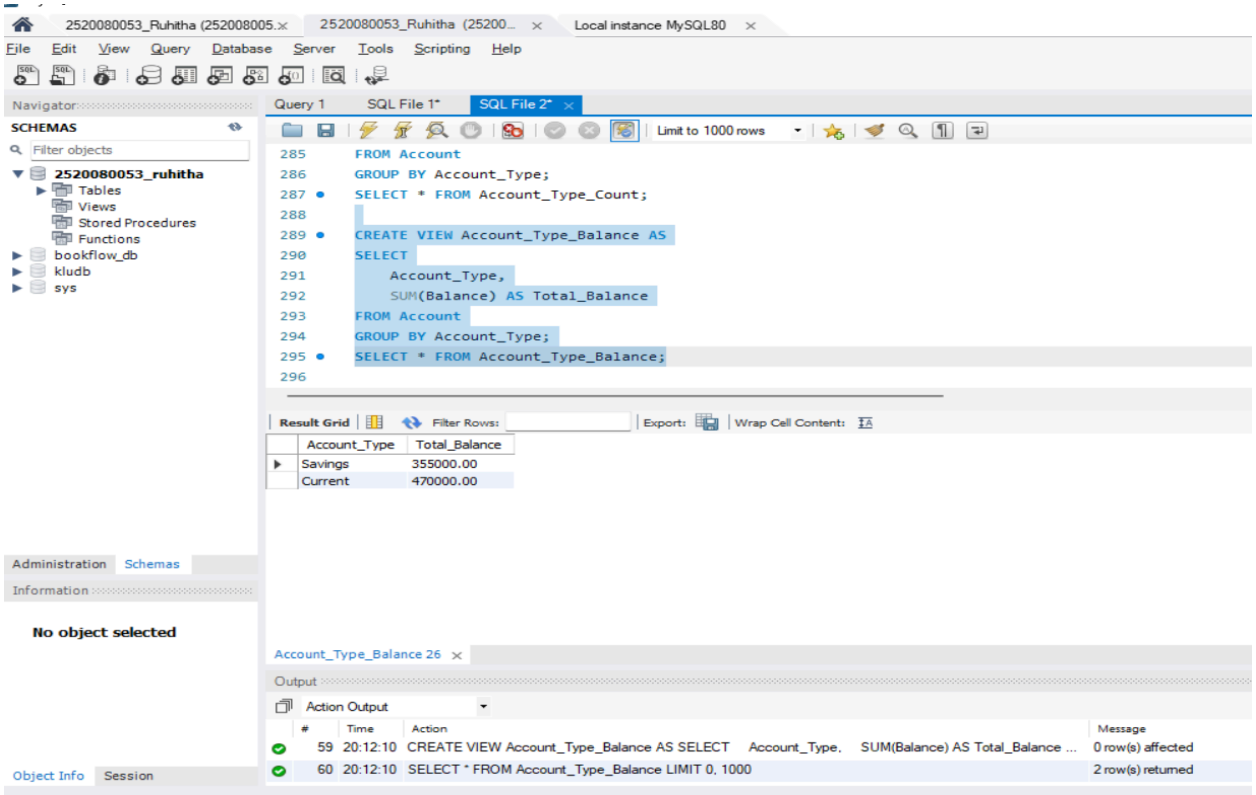
BRANCH-WISE TOTAL BALANCE



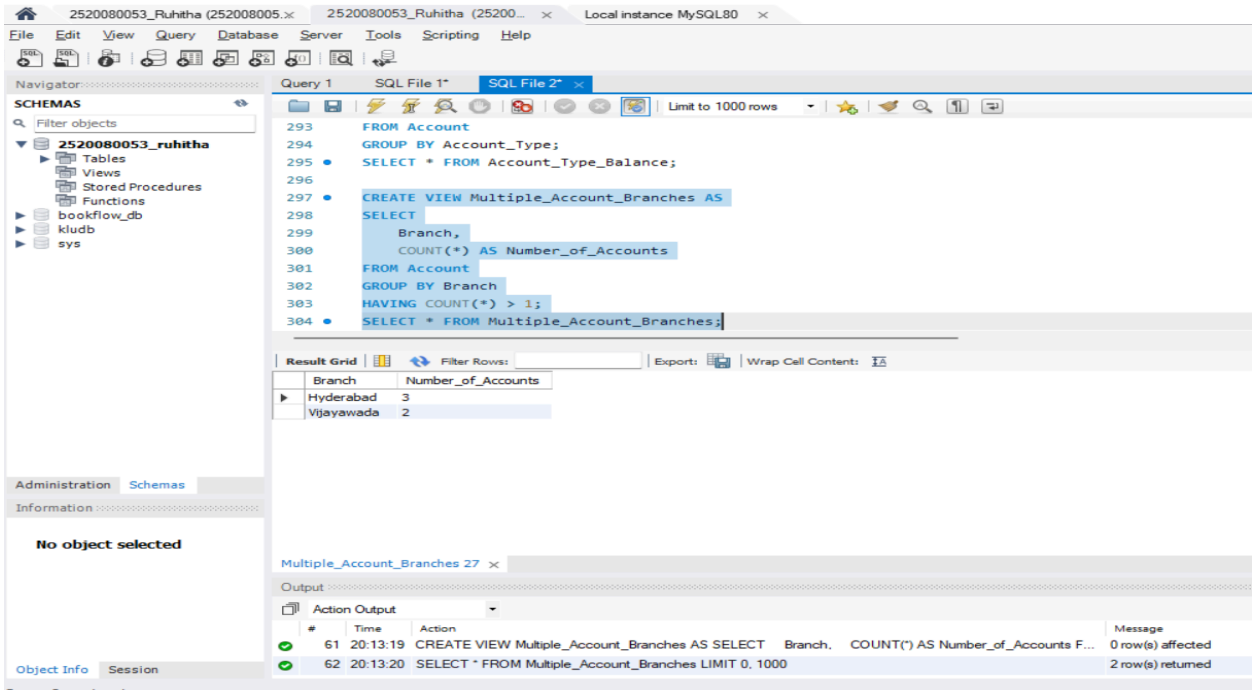
ACCOUNT TYPE COUNT



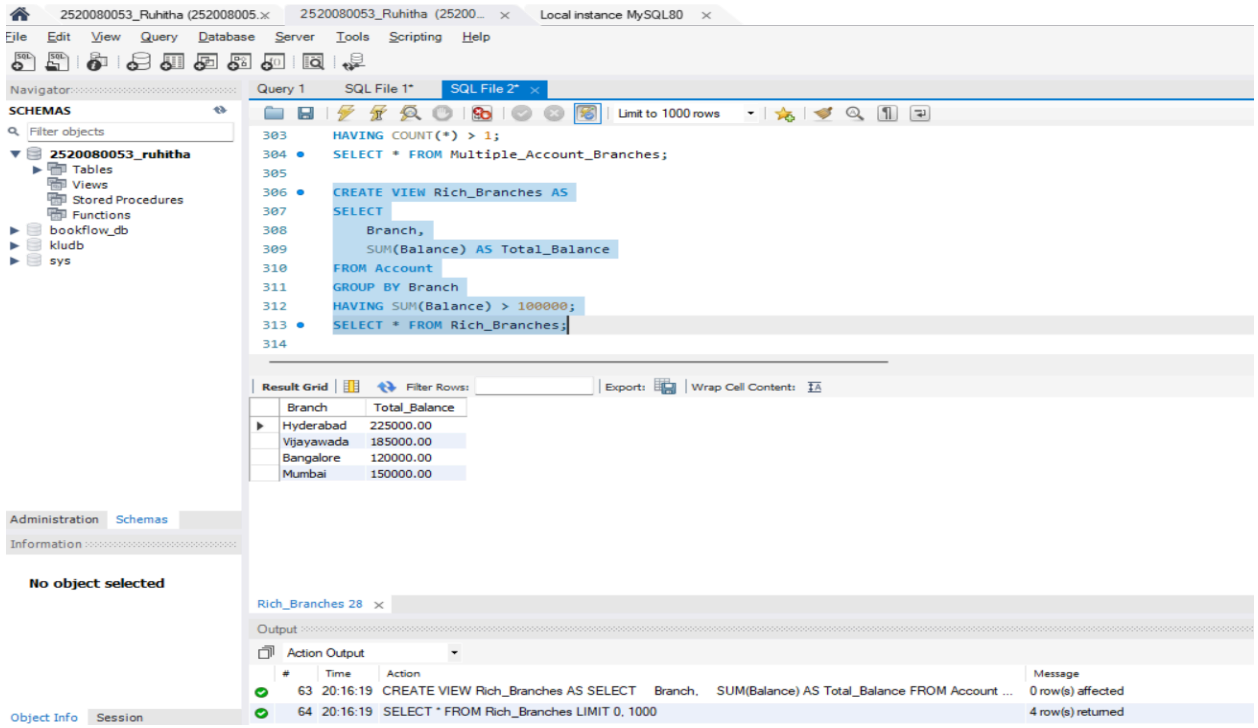
ACCOUNT TYPE TOTAL BALANCE



BRANCHES WITH MORE THAN ONE ACCOUNT



BRANCHES WITH TOTAL BALANCE ABOVE 100000

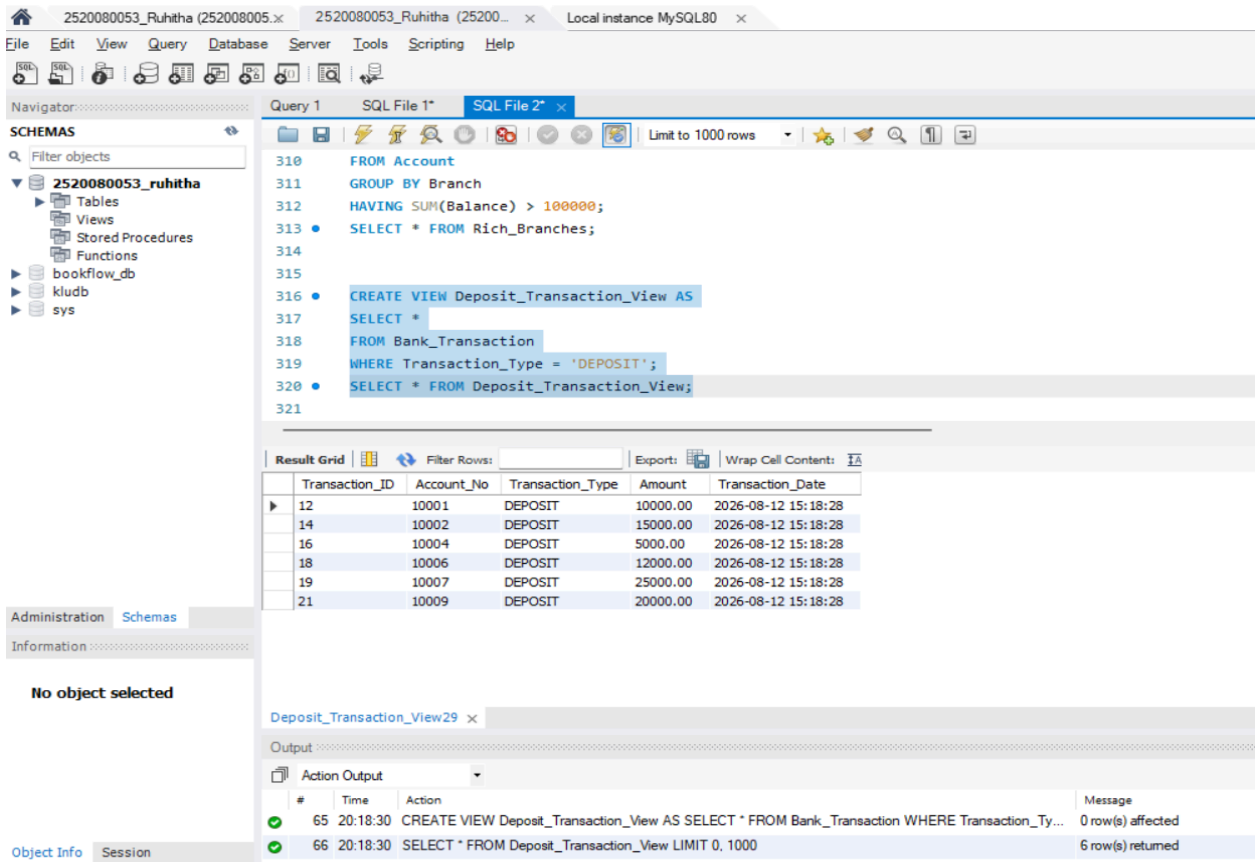


```
HAVING COUNT(*) > 1;
SELECT * FROM Multiple_Account_Branches;

CREATE VIEW Rich_Branches AS
SELECT
    Branch,
    SUM(Balance) AS Total_Balance
FROM Account
GROUP BY Branch
HAVING SUM(Balance) > 100000;
SELECT * FROM Rich_Branches;
```

Branch	Total_Balance
Hyderabad	225000.00
Vijayawada	185000.00
Bangalore	120000.00
Mumbai	150000.00

#	Time	Action	Message
63	20:16:19	CREATE VIEW Rich_Branches AS SELECT Branch, SUM(Balance) AS Total_Balance FROM Account ...	0 row(s) affected
64	20:16:19	SELECT * FROM Rich_Branches LIMIT 0, 1000	4 row(s) returned

DEPOSIT TRANSACTION VIEW

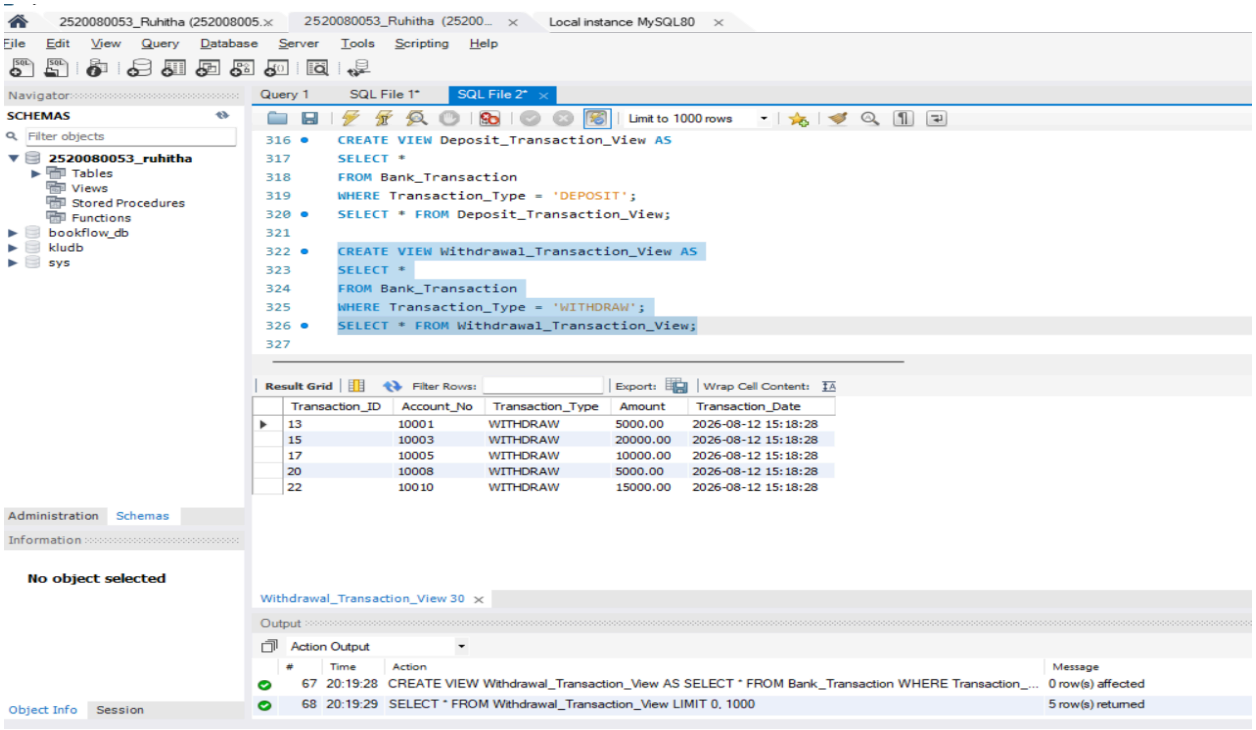


```
FROM Account
GROUP BY Branch
HAVING SUM(Balance) > 100000;
SELECT * FROM Rich_Branches;


CREATE VIEW Deposit_Transaction_View AS
SELECT *
FROM Bank_Transaction
WHERE Transaction_Type = 'DEPOSIT';
SELECT * FROM Deposit_Transaction_View;
```

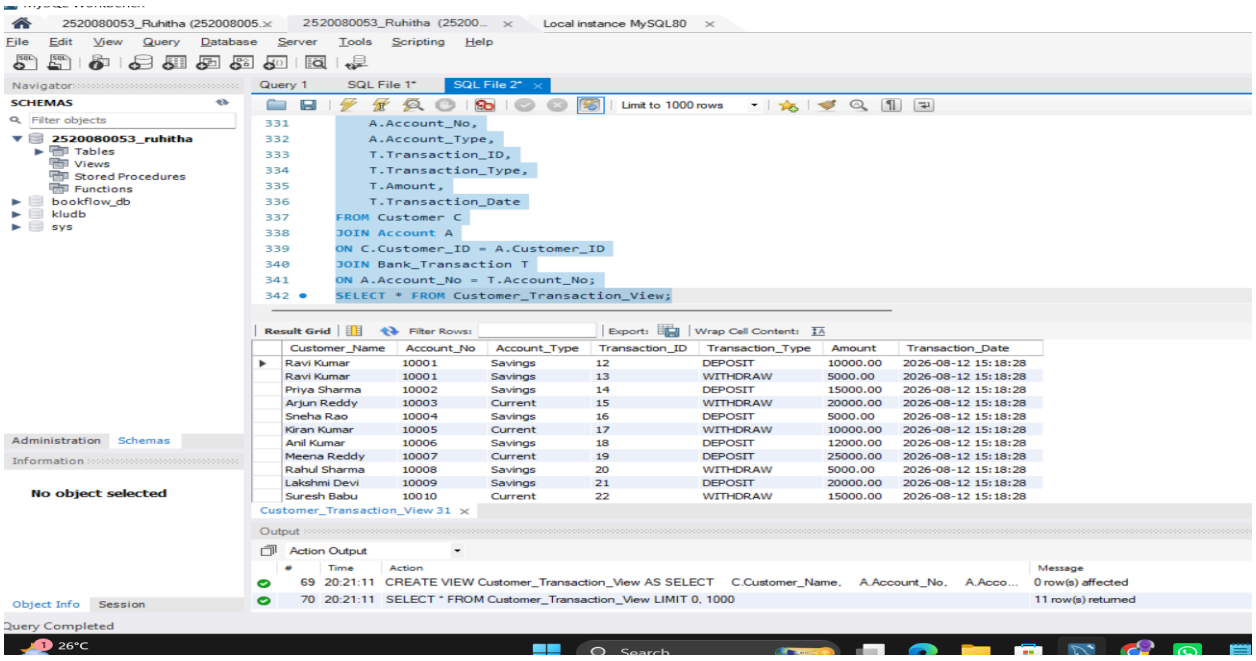
Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
12	10001	DEPOSIT	10000.00	2026-08-12 15:18:28
14	10002	DEPOSIT	15000.00	2026-08-12 15:18:28
16	10004	DEPOSIT	5000.00	2026-08-12 15:18:28
18	10006	DEPOSIT	12000.00	2026-08-12 15:18:28
19	10007	DEPOSIT	25000.00	2026-08-12 15:18:28
21	10009	DEPOSIT	20000.00	2026-08-12 15:18:28

#	Time	Action	Message
65	20:18:30	CREATE VIEW Deposit_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction_Ty...	0 row(s) affected
66	20:18:30	SELECT * FROM Deposit_Transaction_View LIMIT 0, 1000	6 row(s) returned

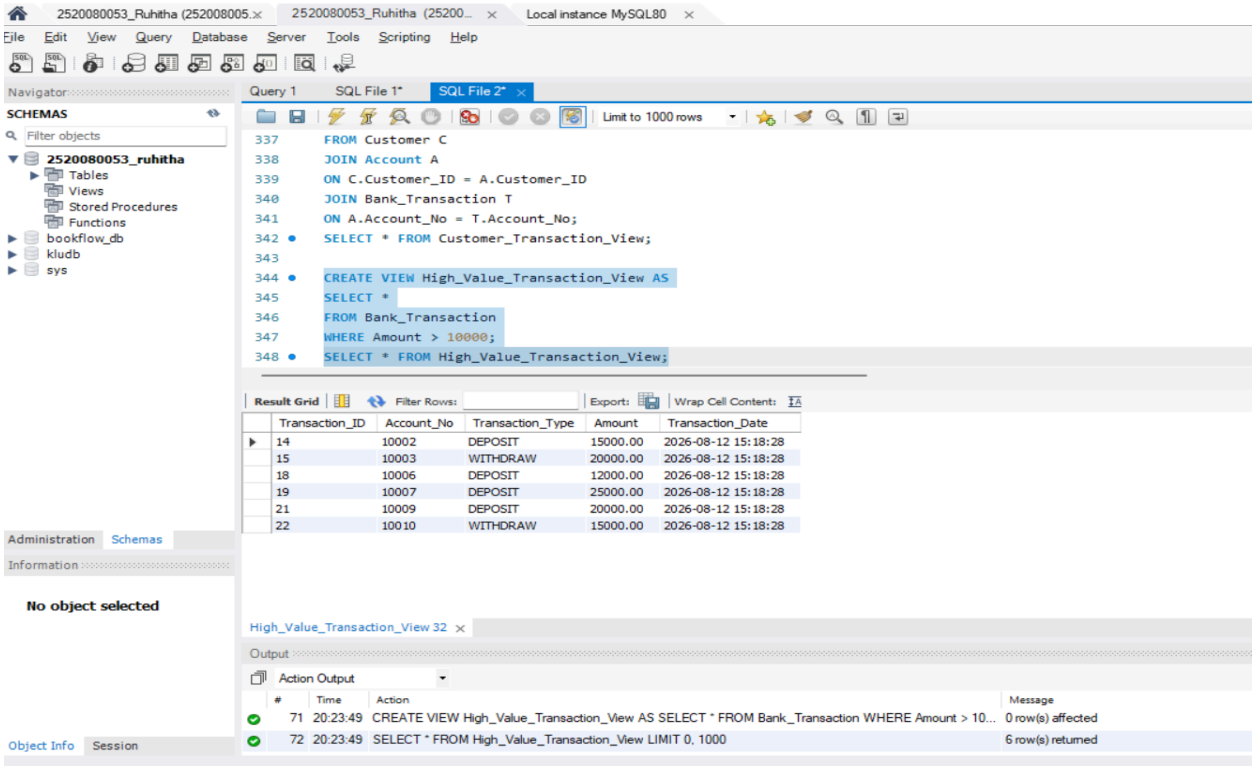
WITHDRAWAL TRANSACTION VIEW



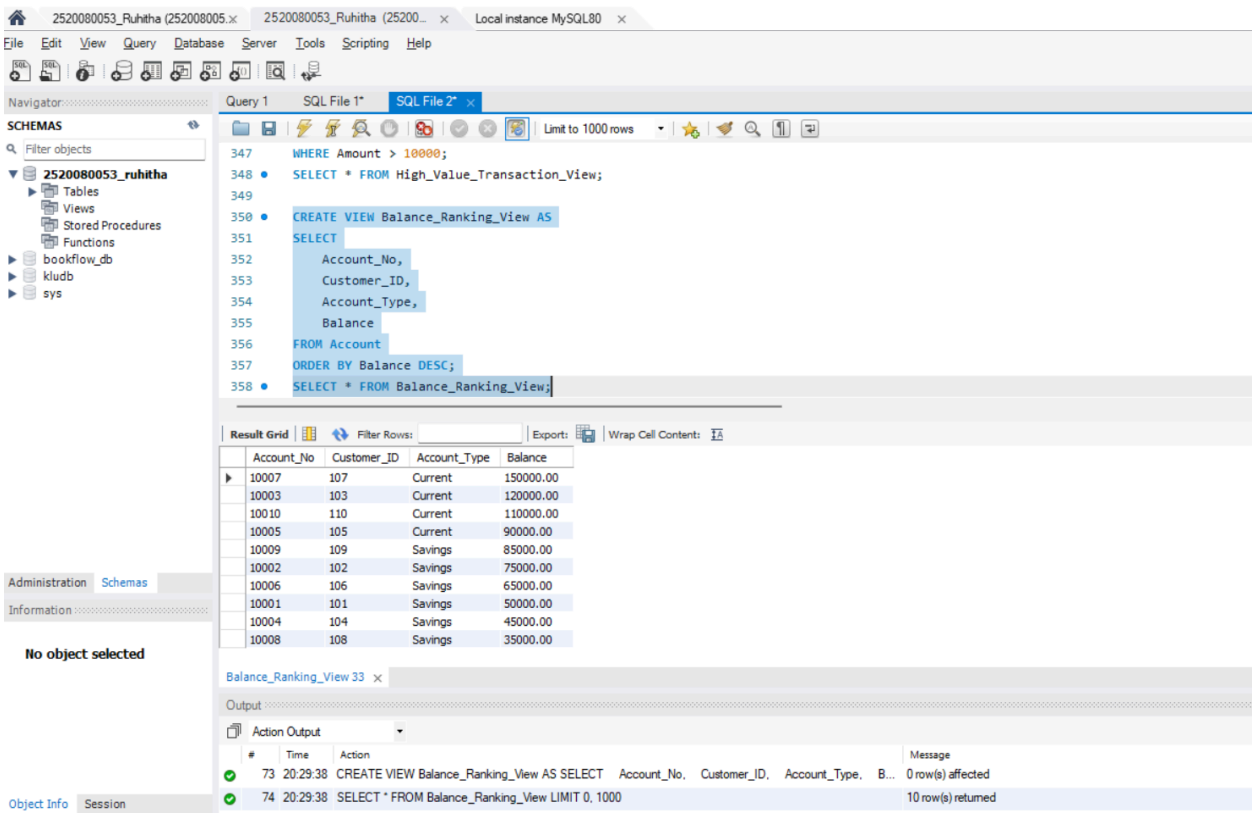
TRANSACTION DETAILS WITH CUSTOMER



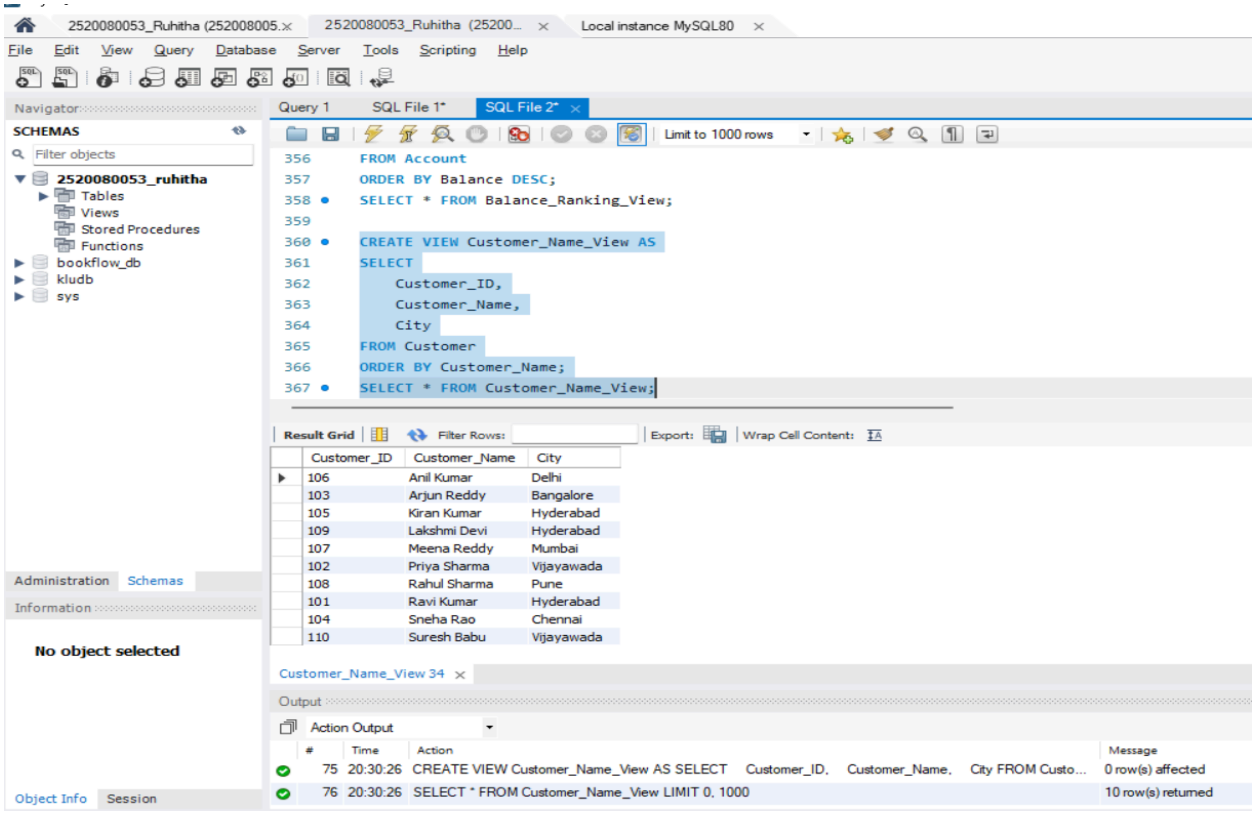
HIGH VALUE TRANSACTION VIEW



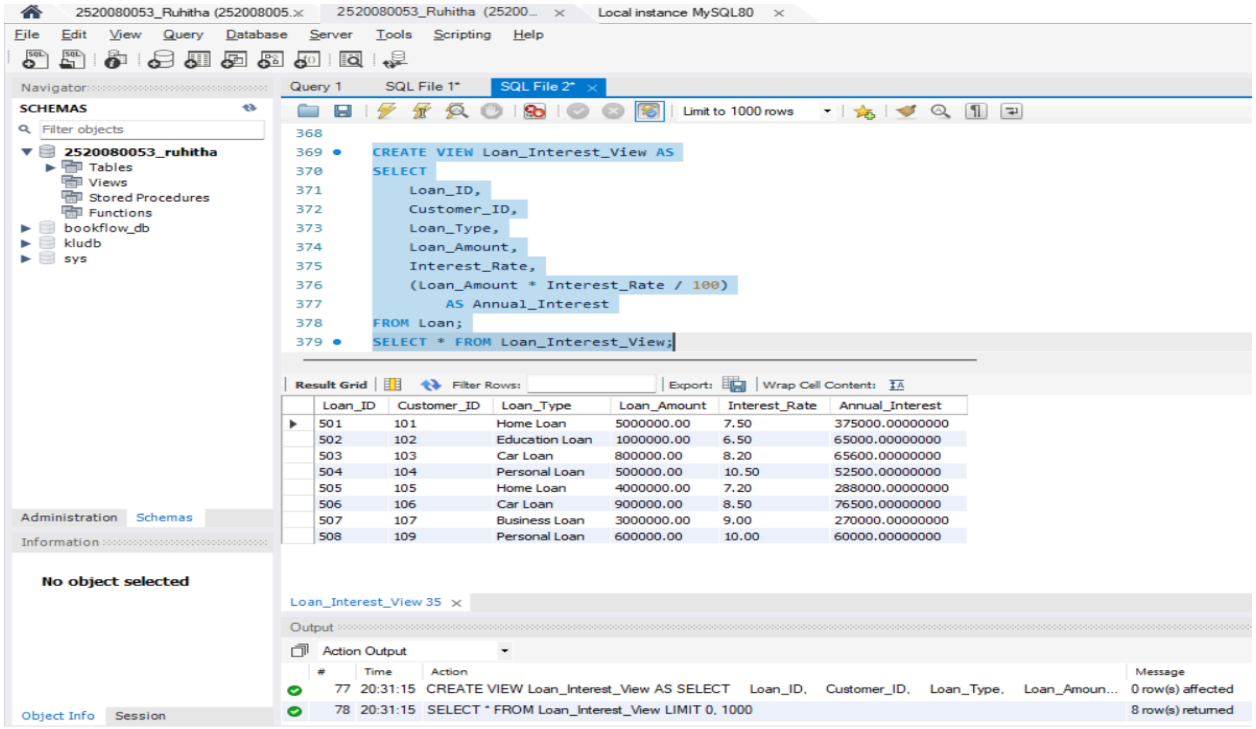
ACCOUNTS ORDERED BY BALANCE



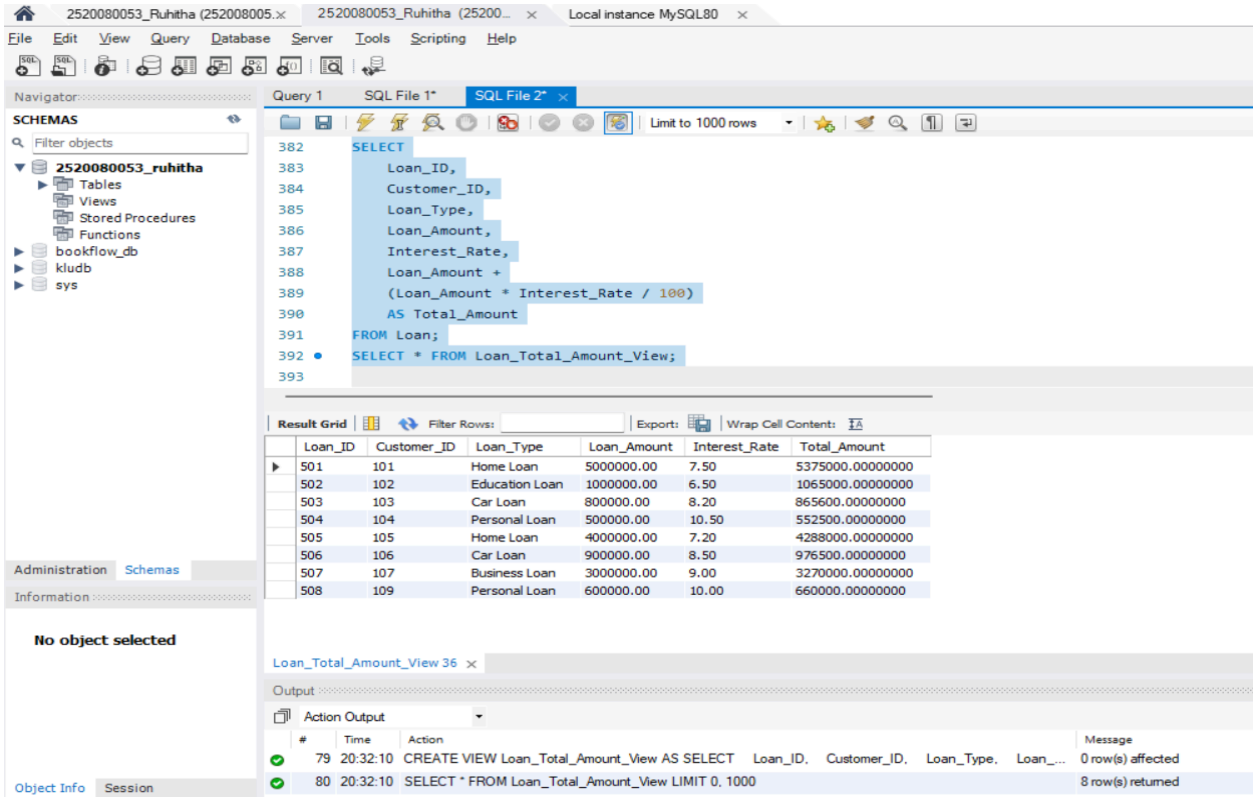
CUSTOMERS ORDERED BY NAME



LOAN INTEREST CALCULATION VIEW



LOAN TOTAL AMOUNT VIEW

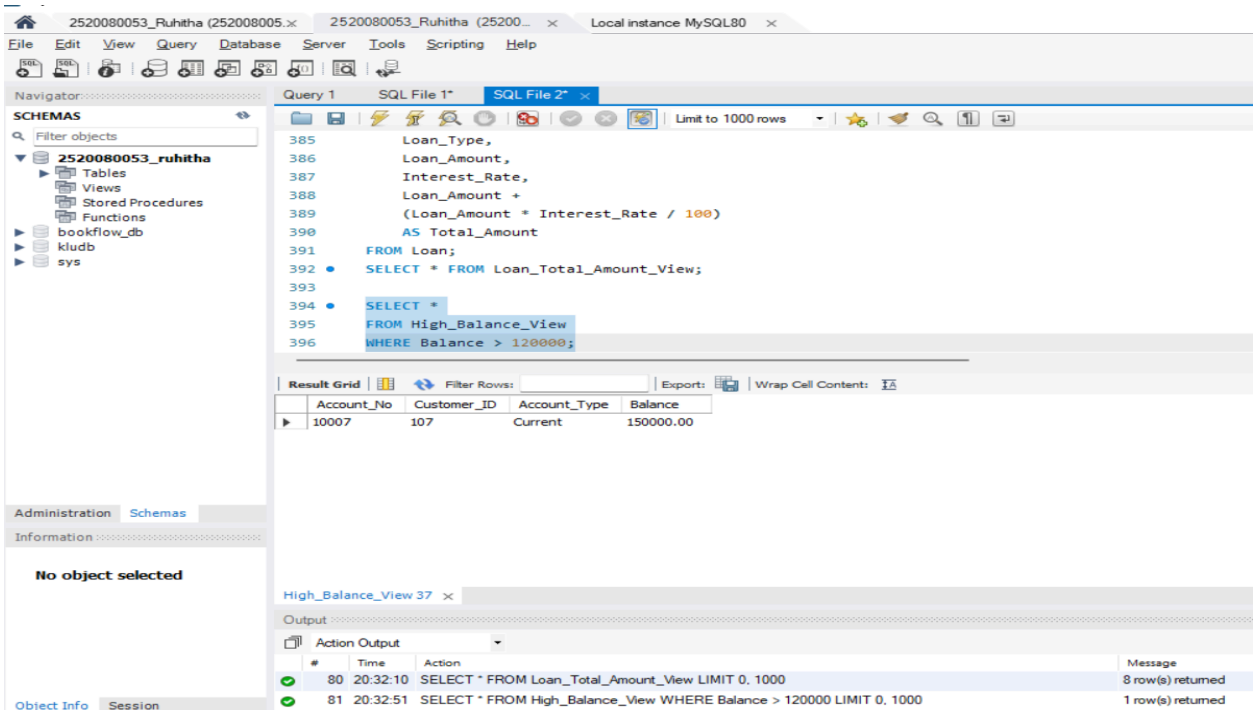


```
SELECT
    Loan_ID,
    Customer_ID,
    Loan_Type,
    Loan_Amount,
    Interest_Rate,
    Loan_Amount +
    (Loan_Amount * Interest_Rate / 100)
    AS Total_Amount
FROM Loan;
SELECT * FROM Loan_Total_Amount_View;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Total_Amount
501	101	Home Loan	5000000.00	7.50	5375000.00000000
502	102	Education Loan	1000000.00	6.50	1065000.00000000
503	103	Car Loan	800000.00	8.20	865600.00000000
504	104	Personal Loan	500000.00	10.50	552500.00000000
505	105	Home Loan	4000000.00	7.20	4288000.00000000
506	106	Car Loan	900000.00	8.50	976500.00000000
507	107	Business Loan	3000000.00	9.00	3270000.00000000
508	109	Personal Loan	600000.00	10.00	660000.00000000

#	Time	Action	Message
79	20:32:10	CREATE VIEW Loan_Total_Amount_View AS SELECT Loan_ID, Customer_ID, Loan_Type, Loan_...	0 row(s) affected
80	20:32:10	SELECT * FROM Loan_Total_Amount_View LIMIT 0, 1000	8 row(s) returned

DISPLAY HIGH BALANCE ACCOUNTS FROM VIEW

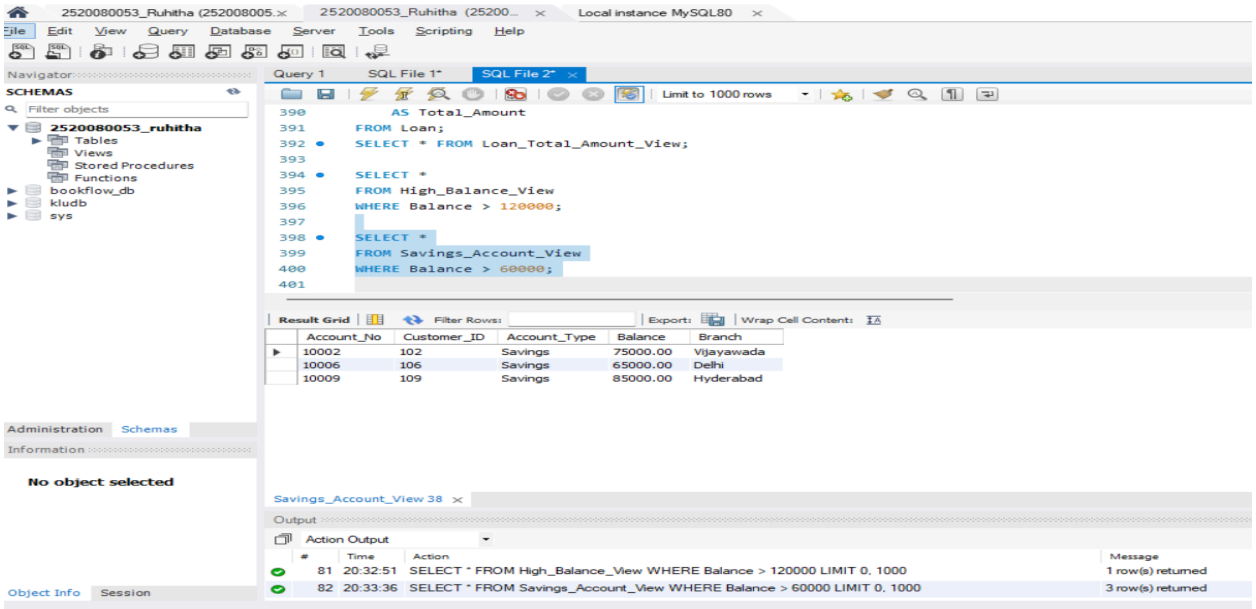


```
SELECT *
FROM High_Balance_View
WHERE Balance > 120000;
```

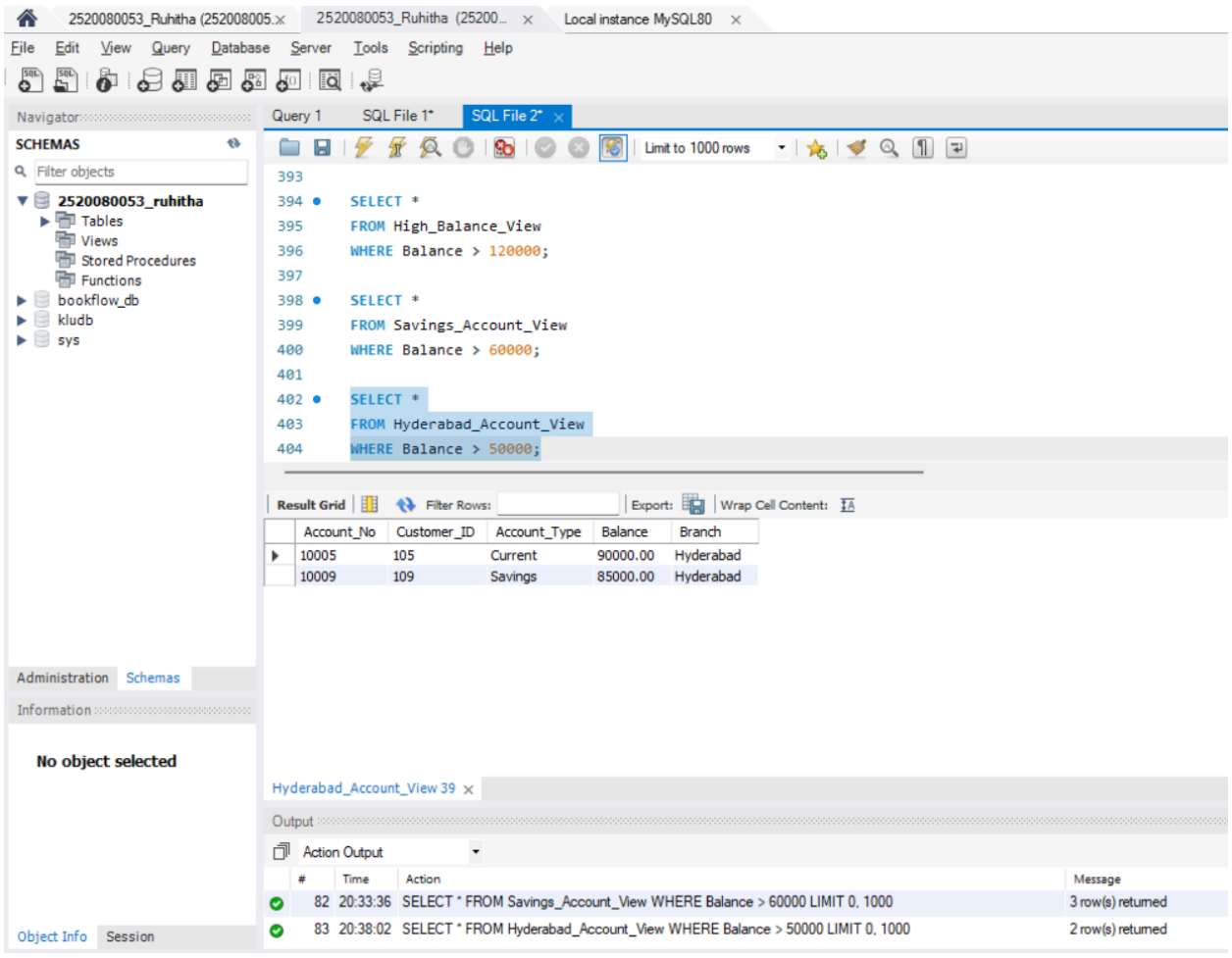
Account_No	Customer_ID	Account_Type	Balance
10007	107	Current	150000.00

#	Time	Action	Message
80	20:32:10	SELECT * FROM Loan_Total_Amount_View LIMIT 0, 1000	8 row(s) returned
81	20:32:51	SELECT * FROM High_Balance_View WHERE Balance > 120000 LIMIT 0, 1000	1 row(s) returned

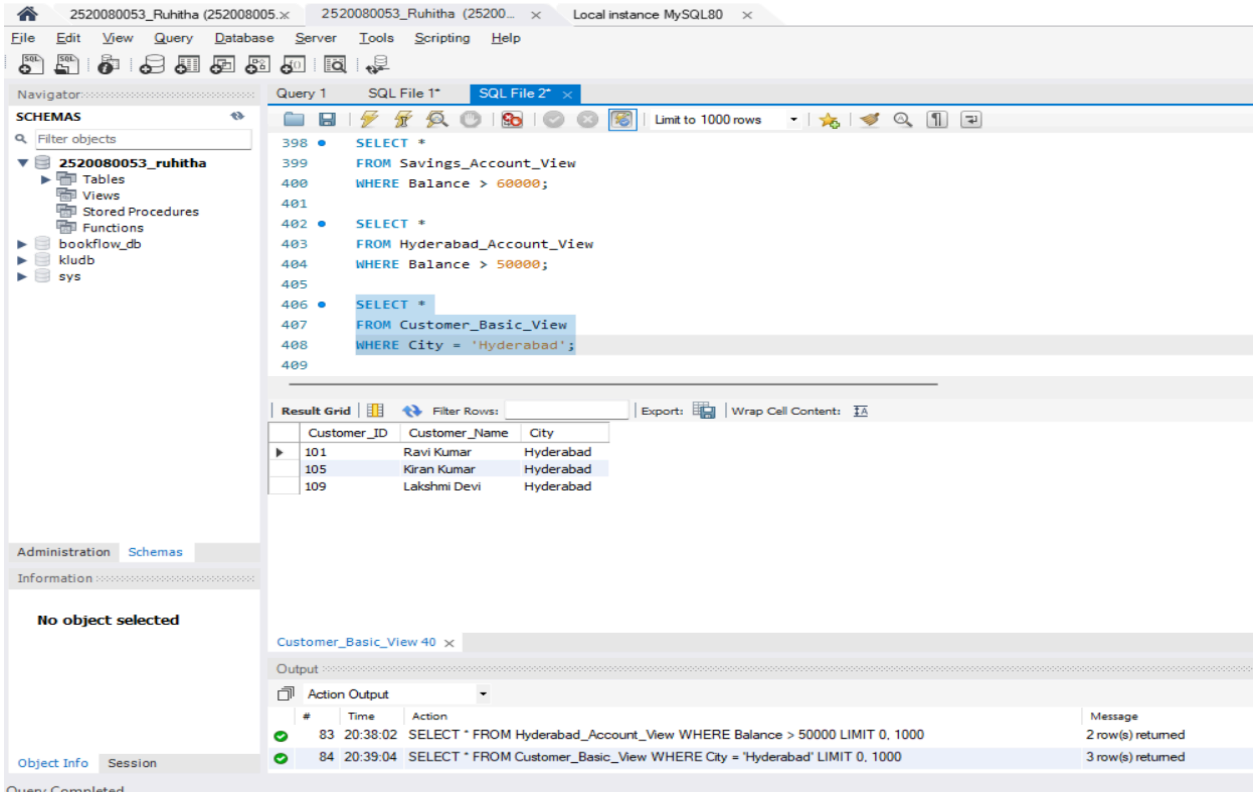
DISPLAY SAVINGS ACCOUNTS ABOVE 60000



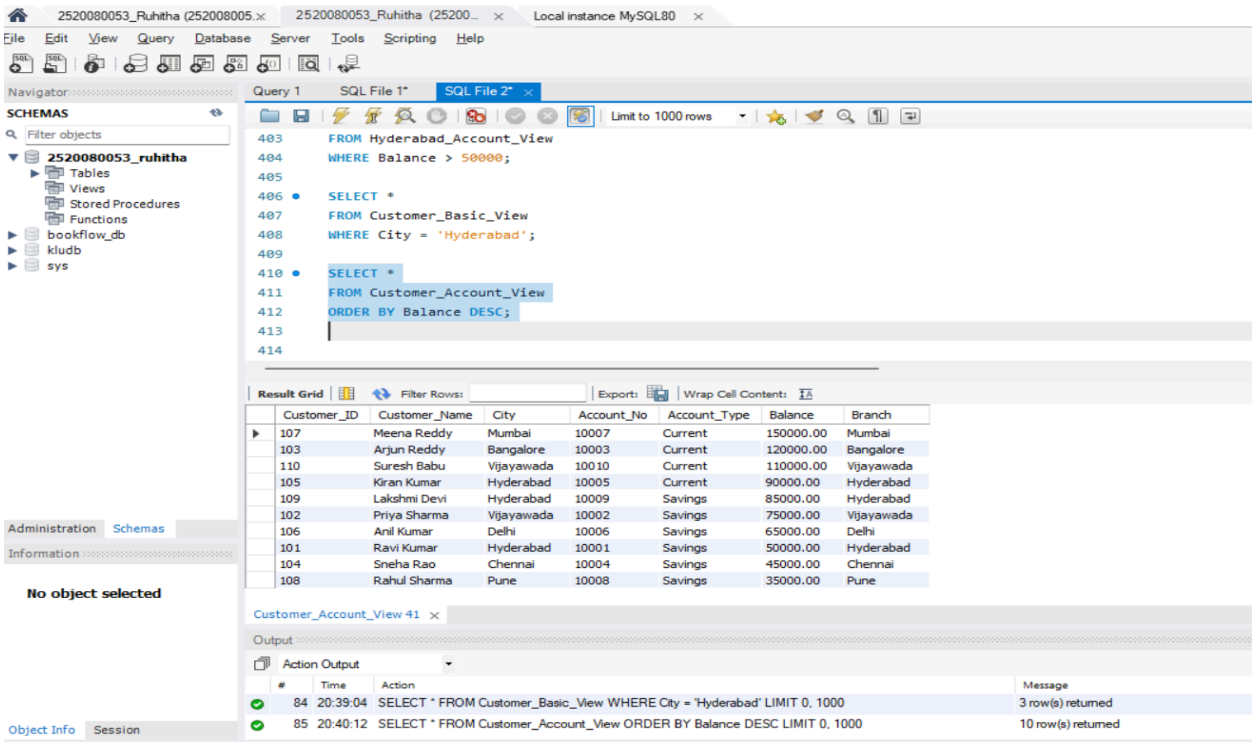
DISPLAY HYDERABAD ACCOUNTS ABOVE 50000



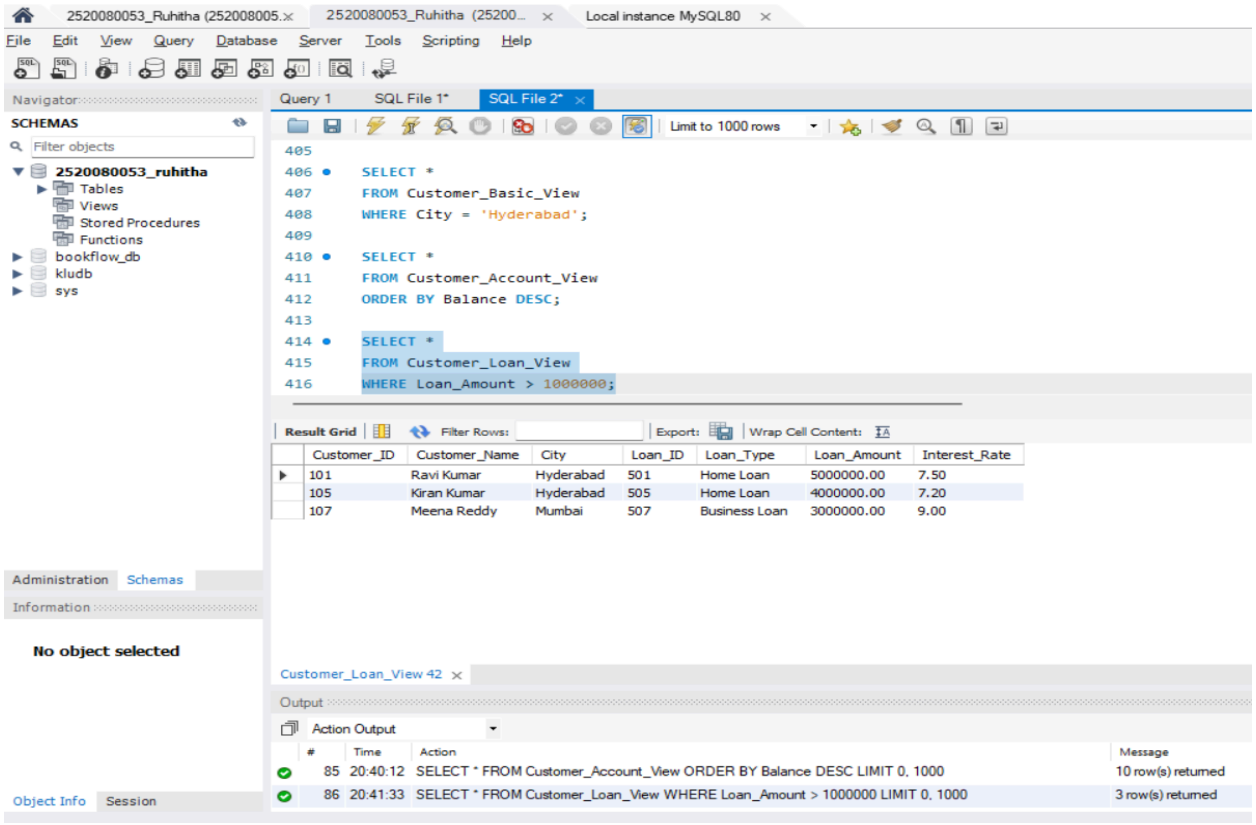
DISPLAY CUSTOMERS FROM HYDERABAD



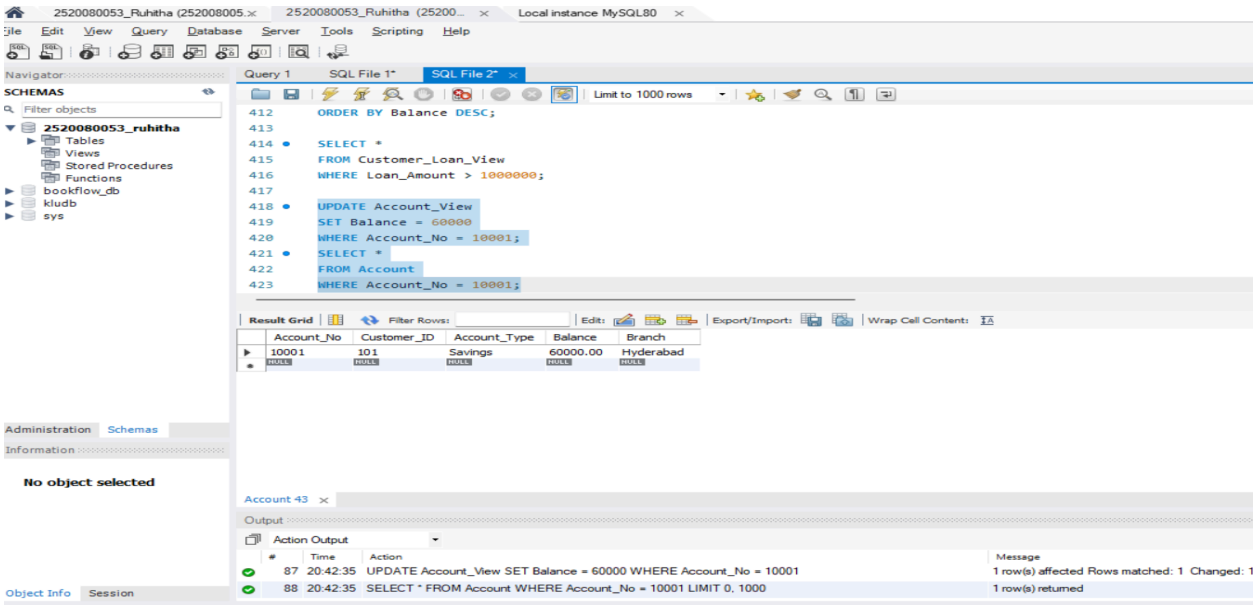
DISPLAY CUSTOMER ACCOUNTS ORDERED BY BALANCE



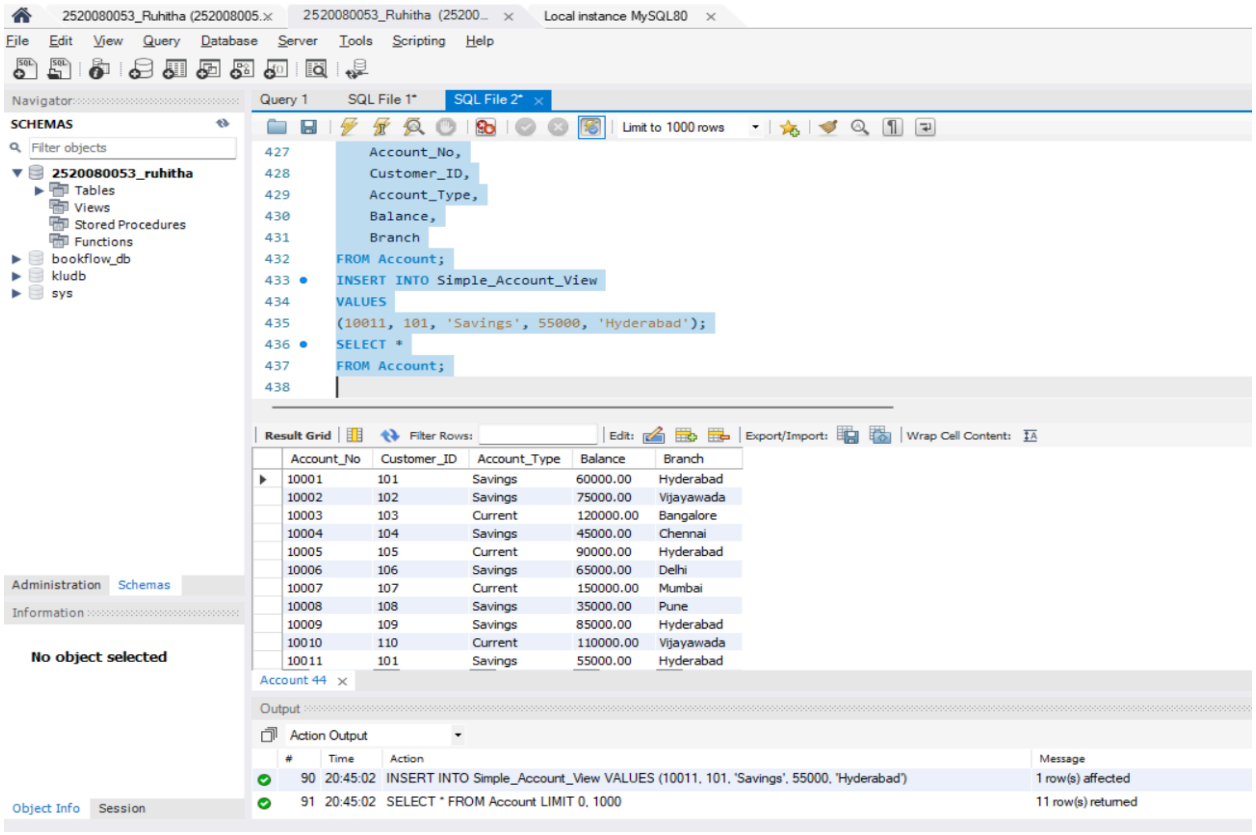
DISPLAY LOANS ABOVE 1000000



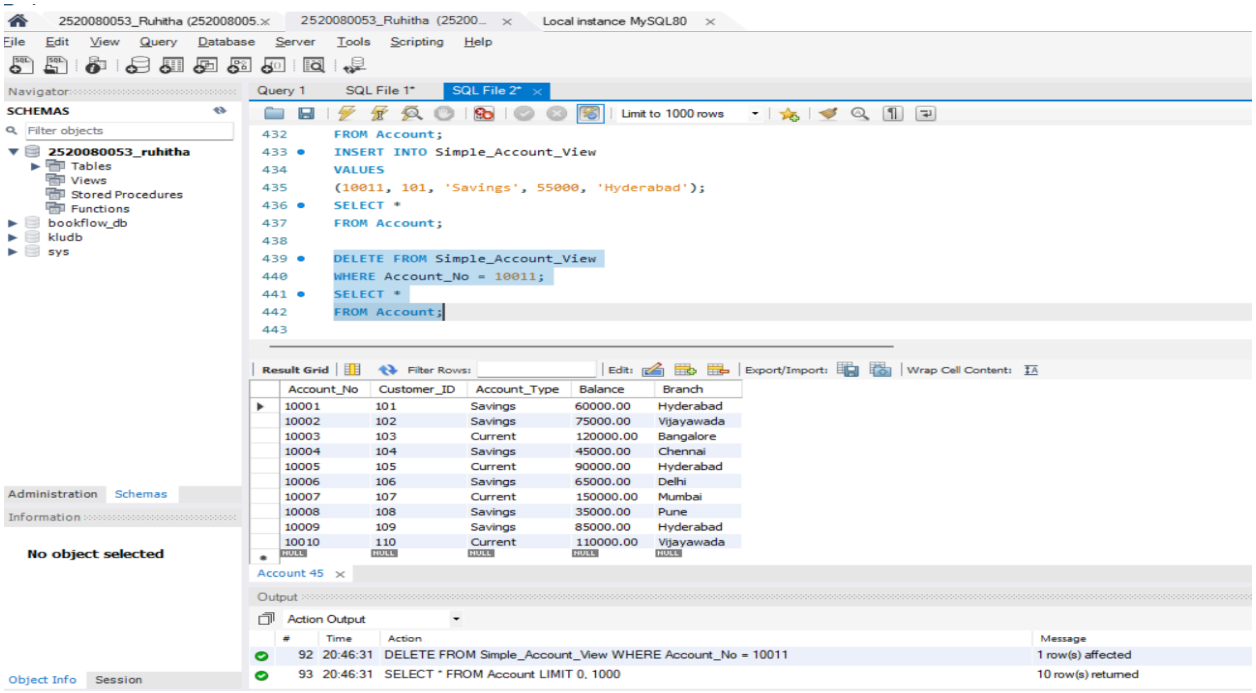
UPDATE ACCOUNT BALANCE THROUGH VIEW



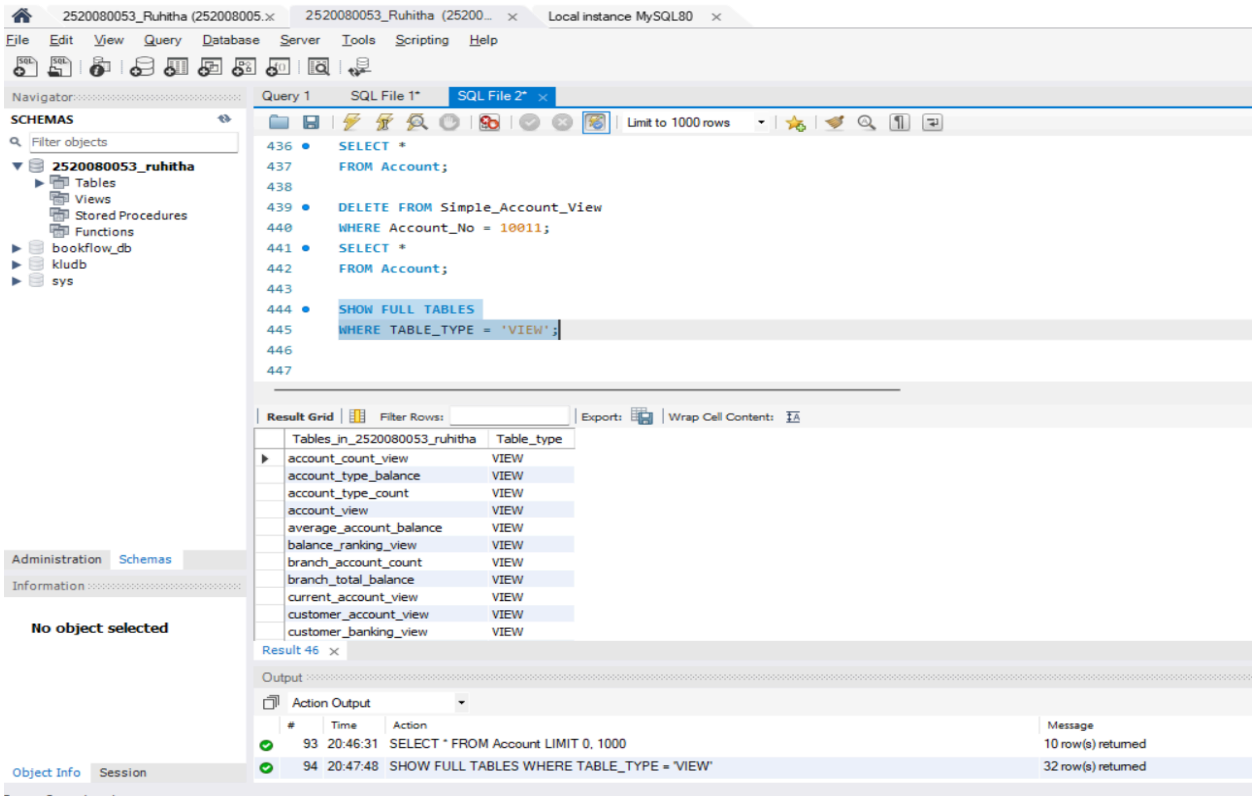
INSERT USING SIMPLE VIEW



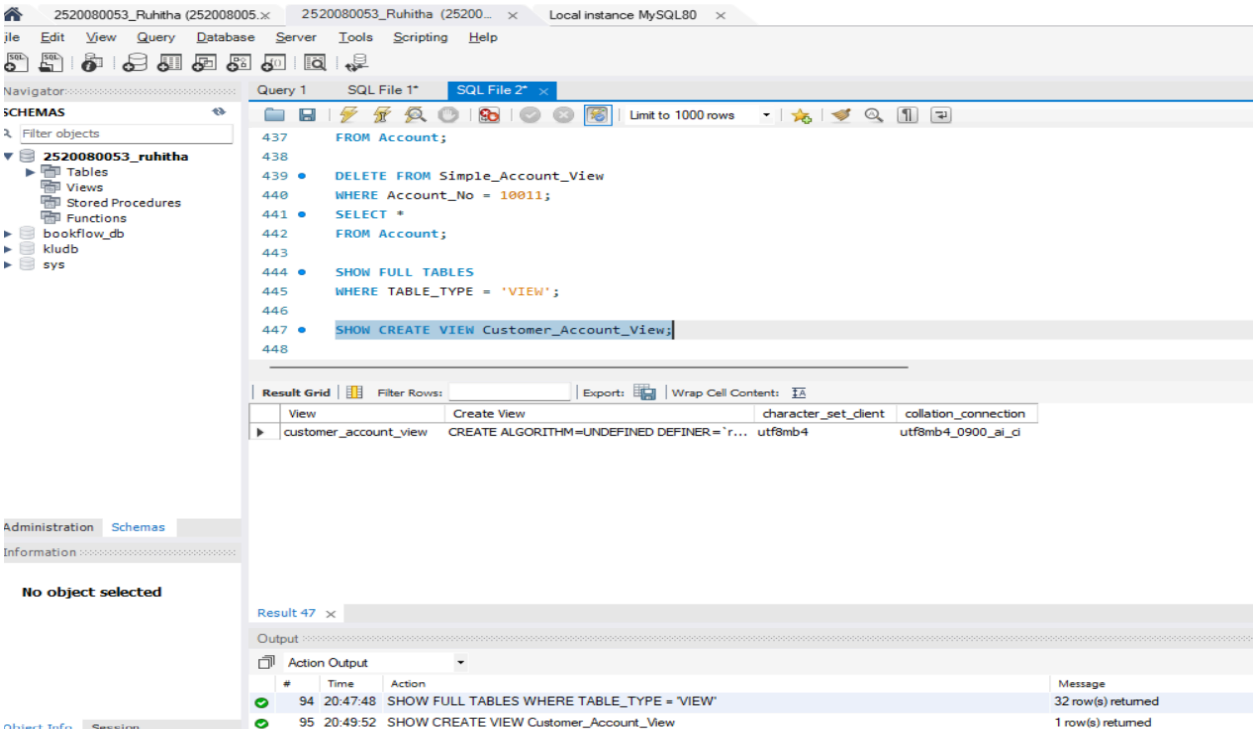
DELETE USING SIMPLE VIEW



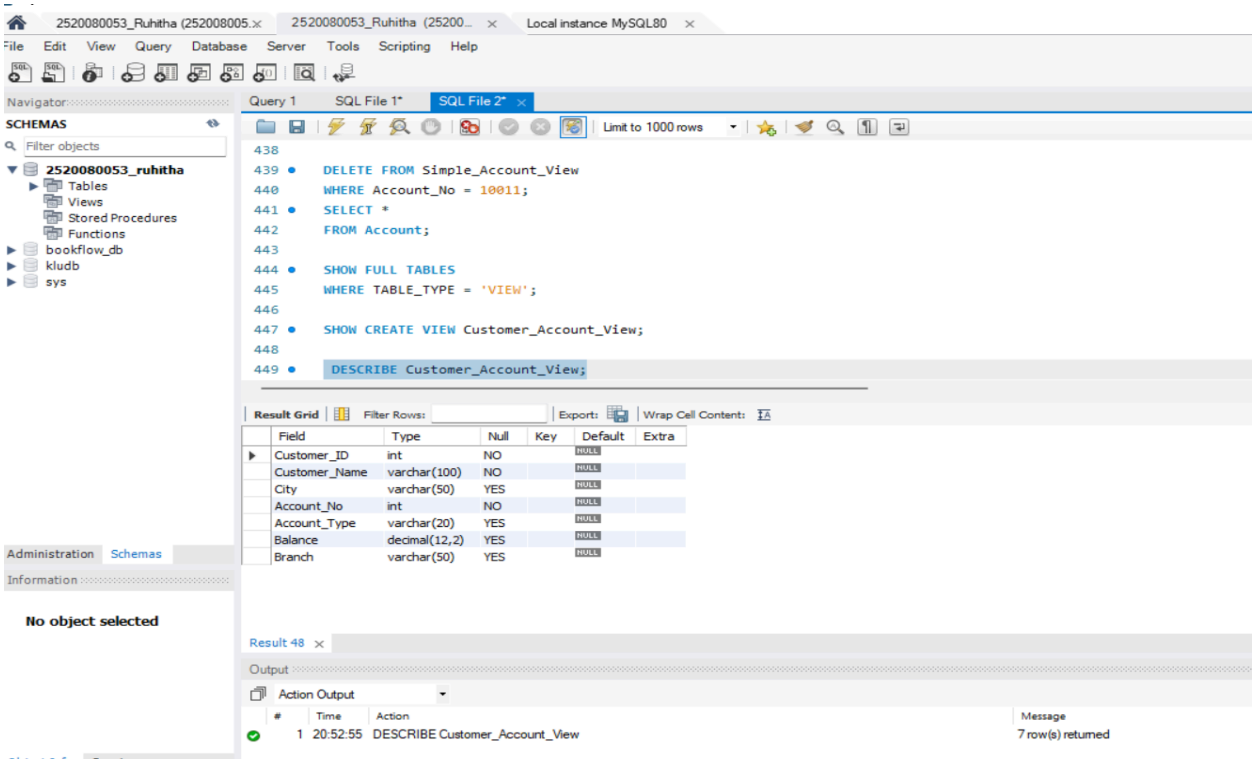
SHOW ALL VIEWS



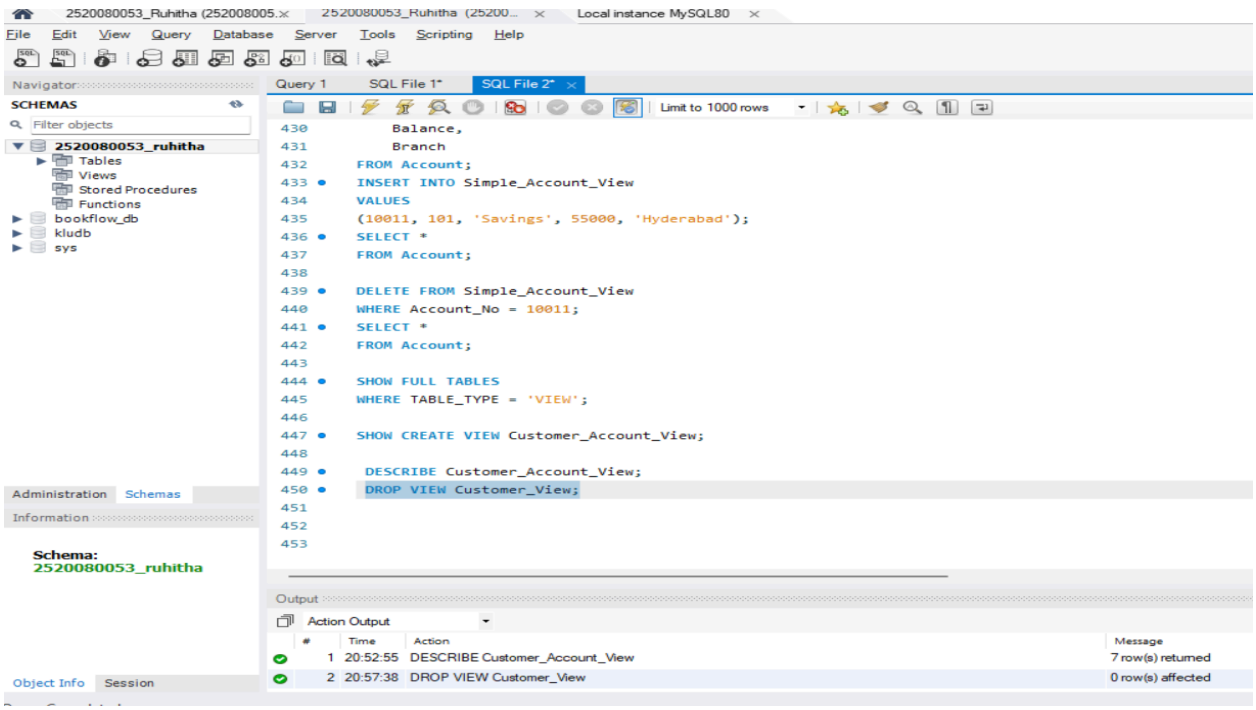
SHOW VIEW DEFINITION



DESCRIBE VIEW



DROP ONE VIEW



DROP MULTIPLE VIEWS